

Element D

Design Concepts, Analysis, and Selection

Problem Statement

Reports indicate that 42,000 American homeowners annually sustain recurring neck, shoulder, and back injuries while picking up and disposing of leaves in their yards. Disregarding this task causes leaves to accumulate in yards, smothering grass and blocking the sunlight needed for proper growth. Overall, leaf removal is a task that can cause physical strain when done and, if neglected, can affect the yard's health, leading to grass death in the fall.

Choosing A Design

Selecting a final design was a multiple-step process to ensure the product met all design requirements that would save time removing leaves and be unique in the market.

The process started with each team member brainstorming 15 unique designs with brief descriptions. The group then came together to sort through the ideas and select the top 6 (See Appendix C), which were put into a decision matrix from the chosen criteria. The top 3 ideas from the decision matrix were then put into a survey for stakeholders to decide on 1 final design.

Brainstorming

From the 45 designs (see Appendix B for all 45 designs) that were created during brainstorming, the group chose the top 6 designs, making sure that they were unique from the market and matched these main design requirements (see Appendix A for all design requirements):

1. Efficiency: The machine shall pick up leaves from a 2,500 sq. ft. lawn within 60 minutes under weather conditions such as clear skies or light wind.
2. Leaf Disposal: The machine shall autonomously pick up leaves and deposit them into a leaf bag
3. Target Cost: The machine shall not exceed the sale cost of \$550
4. Storage: The machine shall have sufficient storage to accommodate at least 30 gallons of leaves with a weight capacity of 50 lbs.
- 5.

The top 6 designs were then put into a decision matrix (see Appendix D for full Decision Matrix):

Criteria	Gulper	Pusher	Auto Fridge	Cart	Lawn Sweeper	Leaf Tank
Cost	4	1	2	3	4	1
Ease of Use	3	4	4	3	3	4
Buildable	5	2	2	4	5	2
Effectiveness	4	3	3	3	4	3
Differ from Market	4	3	4	2	2	3
Storage In Garage	5	3	2	1	3	2
Totals	25	16	17	16	21	15
(1 is the lowest/worst score, 5 is the highest/best score)						

Figure D.1: Product Concept Design Decision Matrix

The criteria used were chosen to ensure that the product was saving time and a viable product for the consumers and for production.

1. Cost:
 - a. Making sure the product is affordable to users by minimizing the material cost needed to build the product without sacrificing its quality
2. Ease of Use:
 - a. The product should be simple and intuitive to use, making it appealing and encouraging for the user to use.
3. Buildable:
 - a. The design must be realistic for the team to build and capable of being tested multiple times.
4. Effectiveness:
 - a. The product should effectively solve the problem by reducing the amount of time and effort it takes to pick up leaves.
5. Differ from Market:
 - a. The product should stand out from current solutions in the market
6. Storage in Garage:
 - a. The product should be compact and space-efficient

The top 3 designs from the decision matrix that were put into the consumer survey were the Auto Fridge, Lawn Sweeper, and Gulper. Below are the descriptions and rough sketches for those designs (see Appendix C for detailed product concept templates for all products in the decision matrix)

1. Gulper

- a. A pull-powered, cubical design with the center of the cube hollowed out and inclined to pick up leaves. The cube has a small incline inside of it to slide the leaves into the reusable leaf bag attached to the cube. The adjustable handle allows the user to pull the Gulper in comfort and directly place the leaves into a bag.

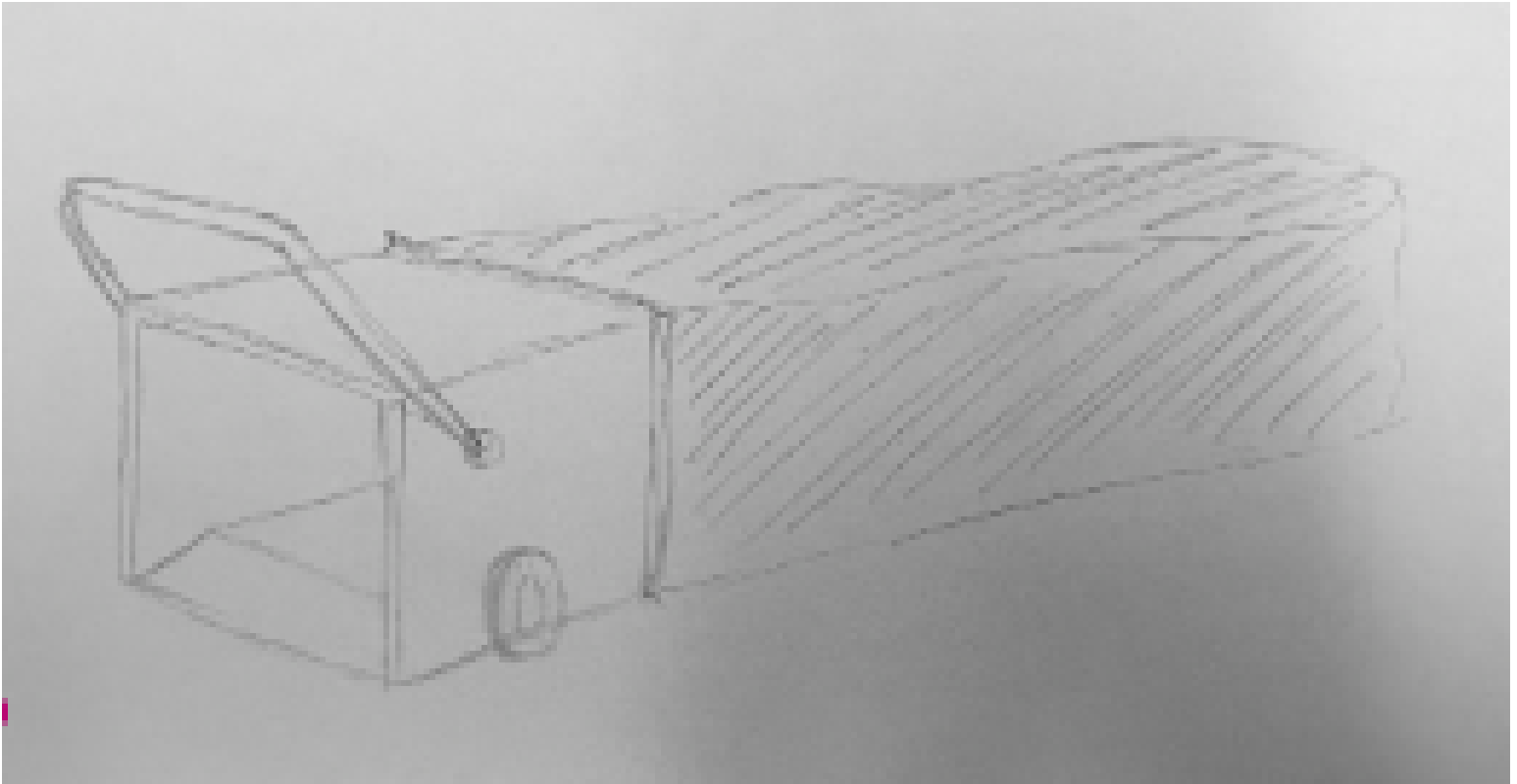


Figure D.2: Leaf gulper concept sketch

2. Auto Fridge

- a. This product is an autonomous electronic leaf-picking machine designed to collect leaves with minimal user effort. It uses an electronic vacuum system to suck leaves up from the ground and stores them in an attached leaf bag. The machine is battery-powered, and the battery is located at the top of the machine for quick replacement and lasts a full cleaning. The product works by following a path preset by the user on an app on their cell phone. The purpose of a replaceable battery is to make sure that the machine can continuously run, and even if one battery dies, another one can be placed in and keep continuous cleaning instead of stopping for recharging.

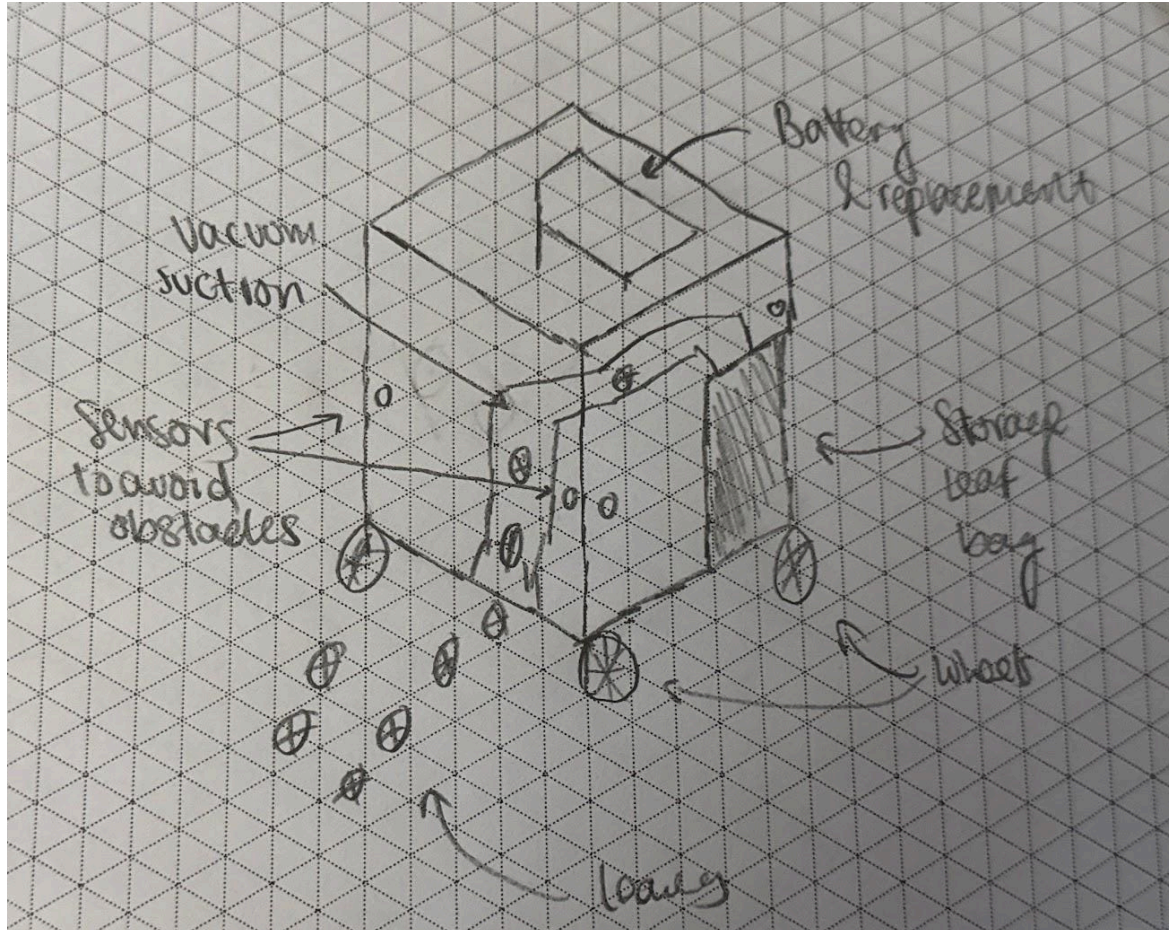


Figure D.3: Auto fridge concept sketch

3. Lawn Sweeper

- a. This product is a manual leaf collector that is pushed and designed to gather leaves without relying on much manual labor. The device has a rotating intake with soft plastic blades that pick up leaves, which are pulled into a heavy-duty bag with a flap for easy disposal. The design is lightweight and durable overall.

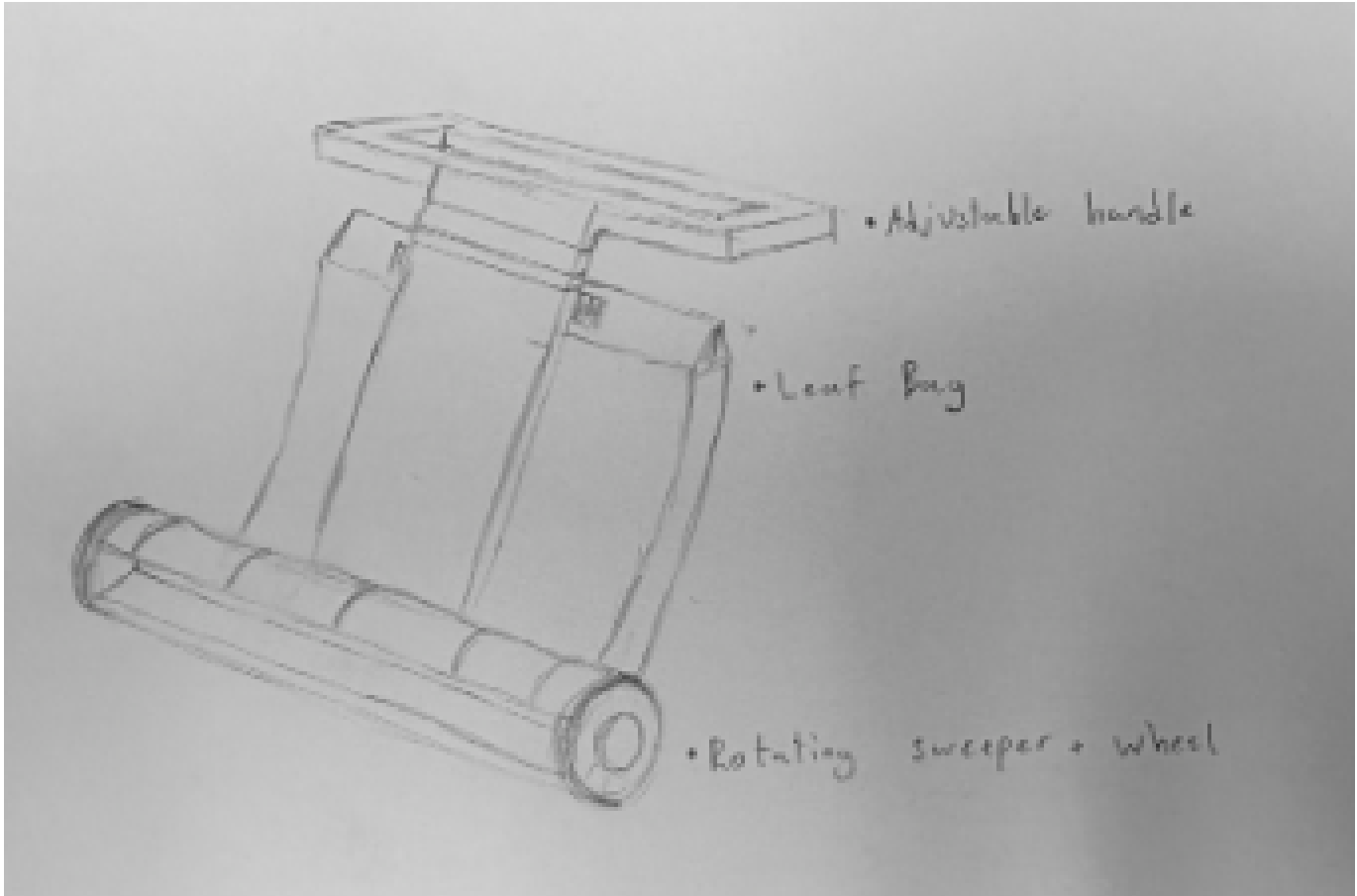


Figure D.4: Lawn sweeper concept sketch

Market Survey

The three designs presented above were put into a market survey to choose a final design, getting 66 responses. The survey was targeted towards stakeholders, distributed across neighborhood HOA groups, homeowners, and teens who work to clean up leaves in the fall season.

The survey presented the three ideas and asked about their impact on the users' current cleanup method and their willingness to use the product.

From the survey data, it was concluded that the Lawn Sweeper was the best design. The rationale behind this conclusion is shown in the data below (See Appendix D for full survey):

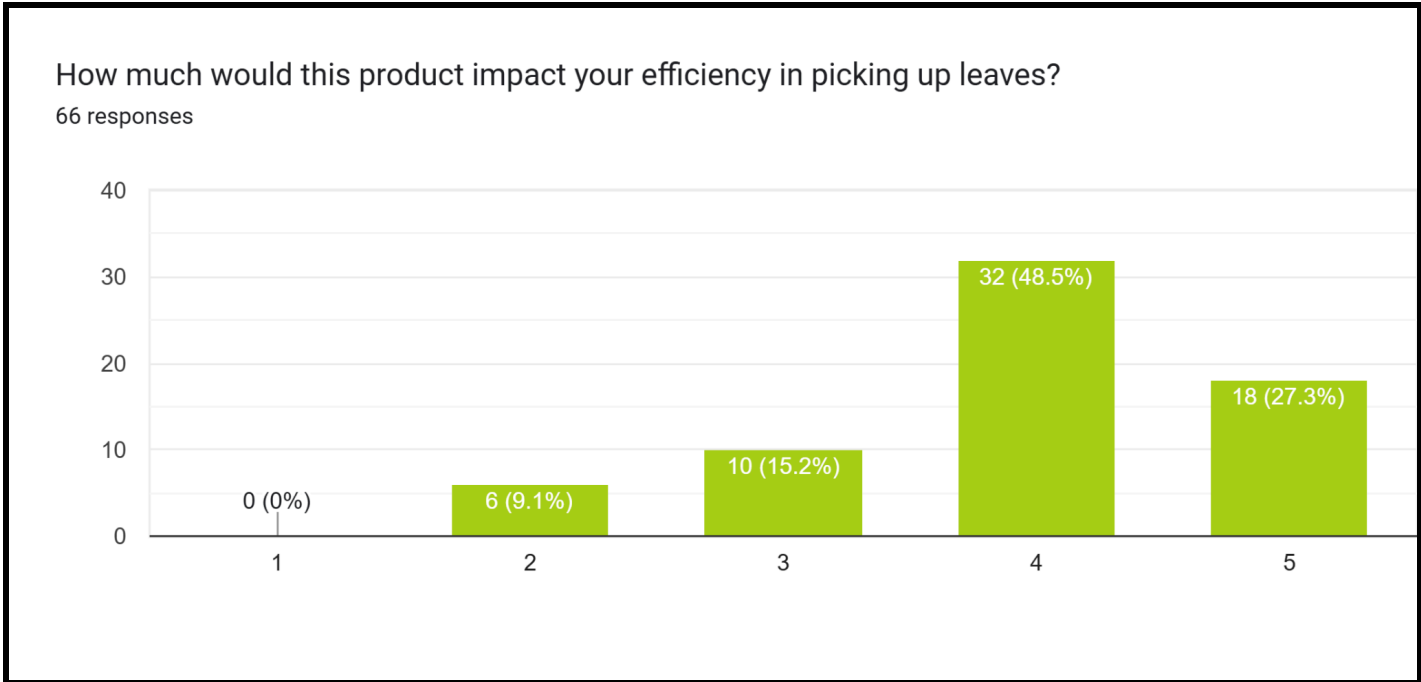


Figure D.5 Survey: Product Efficiency

The question asked in Figure D.5 was asked for each product to assess whether solutions like these would meaningfully impact users and their current leaf-picking methods, or whether they would provide little to no value. Among all three designs, the Lawn Sweeper had the highest average ranking, scoring an average of 4.41, followed by the Auto Fridge and Gulper, having an average ranking of 4.18 and 3.95, respectively.

This gave insight that the Lawn Sweeper is the product that is going to help the user the most in terms of improving their work efficiency, hence reducing the time it takes to work. However, the average efficiency ranking for the auto fridge was not far behind, so it is still a product that is under consideration.

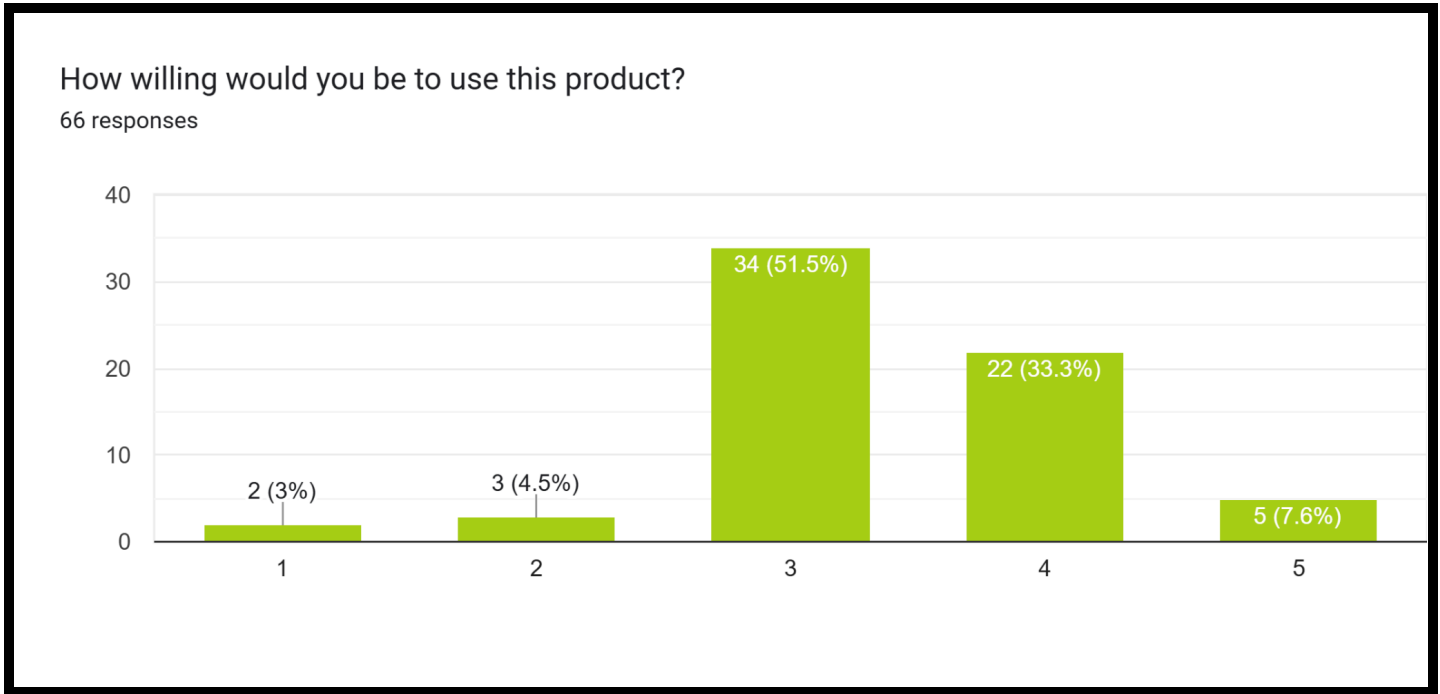


Figure D.6 Survey: Willingness to use product

The question asked in Figure D.6 was also asked for each product to assess whether the user would be willing to use the product if available. Among all three designs, the Lawn Sweeper had the highest average ranking again, scoring an average of 3.52, followed by the Auto Fridge and Gulper, having an average ranking of 3.18 and 3.39, respectively.

This question also ranks the Lawn Sweeper as the best in terms of the willingness of the user to use this product. However, the rankings of all three products are very close, so no definite conclusions can be drawn from these results, but Lawn Sweeper remains the best so far.

Which type of design would you prefer in a product solving this problem

67 responses

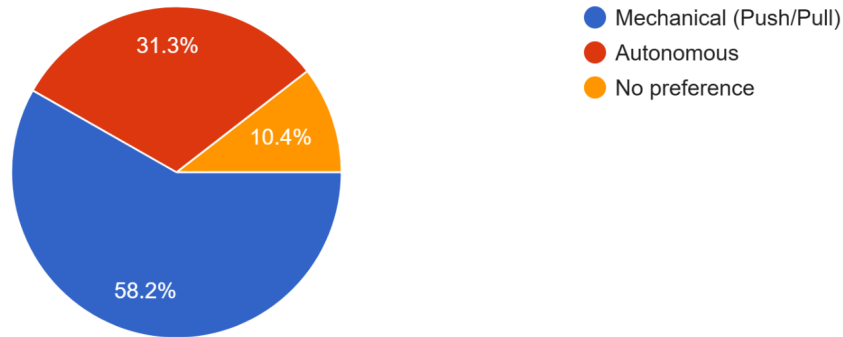


Figure D.7 Survey Design type preference

The question asked in Figure D.7 was to assess which type of design the user would prefer in general, an autonomous one or a mechanical one. From the results of this question, it is clear that the users prefer a mechanical push/pull design versus an autonomous one. The autonomous one had a 31.3% preference, and the mechanical one had 58.2% preference.

This question shows that most users would prefer a design like the Lawn Sweeper or the Gulper over the Auto Fridge, as those two are the mechanical designs, and the Auto Fridge is an autonomous design.

Final Design:

From the results of the survey, it was clear that the Lawn Sweeper emerges as the top product. Next, there was a CAD model developed of the final design of the Lawn Sweeper before it could be built. Below are some main schematic drawings of the product (See Appendix F for full sketches)

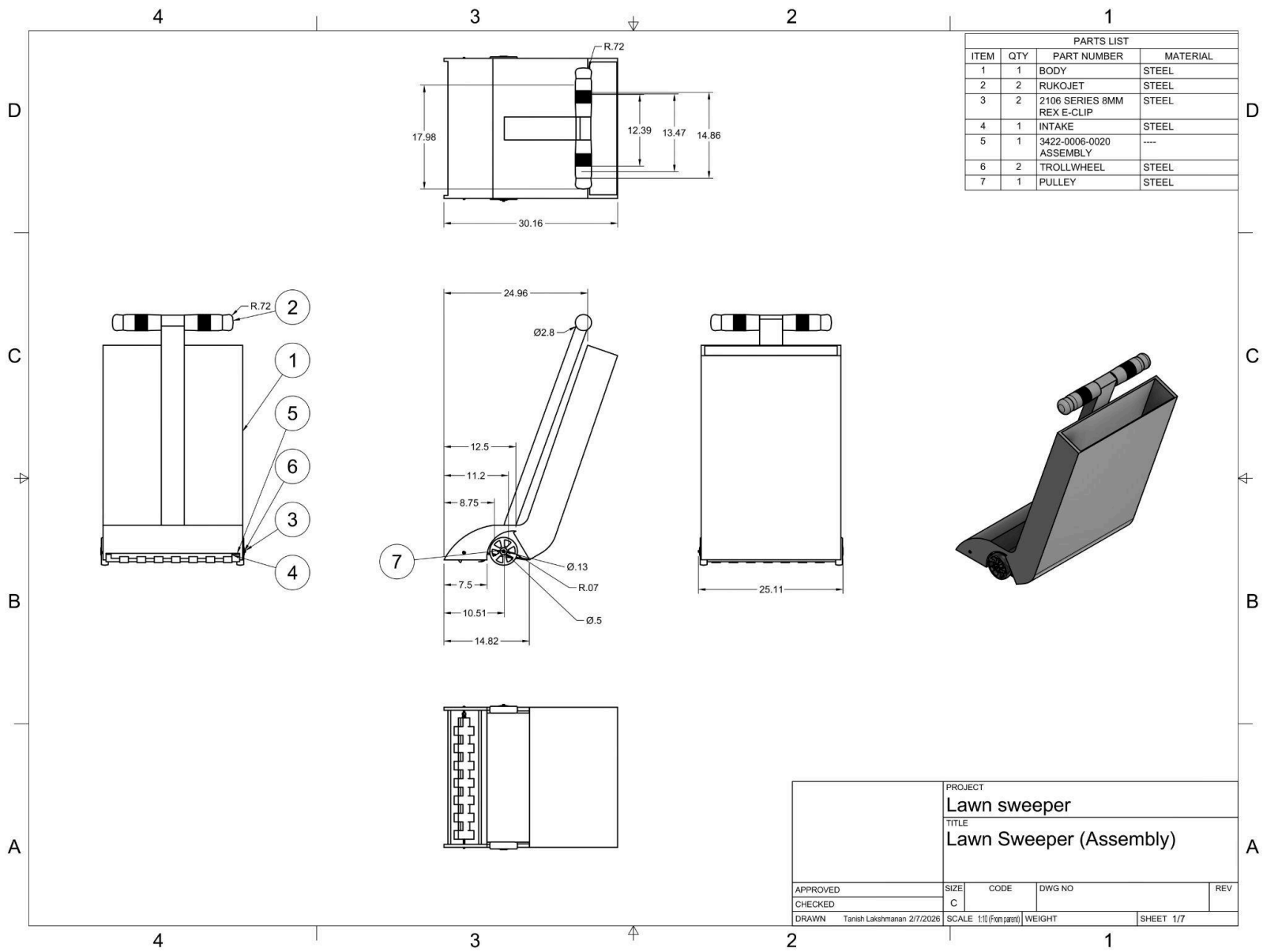


Figure D.8 Assembled Lawn Sweeper

PARTS LIST			
ITEM	QTY	PART NUMBER	MATERIAL
1	1	BODY	STEEL
2	2	RUKOJET	STEEL
3	2	2106 SERIES 8MM REX E-CLIP	STEEL
4	1	INTAKE	STEEL
5	1	3422-0006-0020 ASSEMBLY	----
6	2	TROLLWHEEL	STEEL
7	1	PULLEY	STEEL

Figure D.9 Full Parts List for Lawn Sweeper

Figure D.9 shows the full parts list for the Lawn Sweeper, most of which materials will be purchased or could be 3D-printed with a material change.

Product Description:

This product is a manual leaf collector that is pushed and designed to gather leaves without relying on much manual labor. The device has a rotating intake with soft plastic blades that pick up leaves, which are pulled into a heavy-duty bag with a flap for easy disposal. The design is lightweight and durable overall.

Operation:

The input is human force. When the user pushes the machine forward, it provides the machine with mechanical power to spin the intake rotator at the base. When the intake comes in contact with the leaves, it lifts them up, and the internal mechanisms guide the leaves into a leaf bag for storage. The leaf bag is attached using clips and ensures easy replacement after it is full to minimize physical strain.

Justification:

The problem is that leaf collection methods like raking cause too much physical strain on the user and result in various back and neck injuries because these methods require repeated bending. This product directly works against the problem by having a mechanism that picks up the leaves using mechanical power and minimizes human effort. The handle that the user has to push is adjustable, allowing users to adjust it to their desired setting to ensure an upright position when they clean up their lawn.

This design was chosen because it has the perfect balance between effectiveness, simplicity, and usability. This machine is very effective and does not require electricity or fuel to operate, which reduces the complexity of the product and ensures easier maintenance as well. The usability of this product is also better than current solutions because it keeps the user in a comfortable position the entire time that they are working, preventing injury and reducing tiredness at the end. Many alternative solutions use electricity or fuel sources for power, which increases the complexity of the product and makes it harder for the user to use, as well as increasing noise and costs.

This design also meets many of the design specifications, only lacking in the component of being autonomous, which would require complex electronics and sensors to be included, which increases maintenance and chances of the machine breaking down.

Appendices

Appendix A: Design Requirements

Design requirements below are listed to appeal directly to the stakeholders for leaf removal. Directed toward homeowners and lawn services, the lawn sweeper will be marketed as a more effective product while requiring less maintenance to take care of.

Design Requirements

Below are the design requirements for the lawn sweepers' final design. They are ranked, 1 being the most important and 16 being the least.

1. Efficiency
 - a. Specification:
 - i. The machine shall pick up leaves from a 2,500 sq. ft. lawn within 60 minutes under weather conditions such as clear sky or some wind.
 - b. Rationale:
 - i. This ensures that the machine performs at a speed comparable to manual raking or blowing, making the product a practical solution for everyday use. Typically, manually raking a 2,500 sq. ft lot takes one person 2-3 hours to complete¹, much longer than the targeted time for the machine.
2. Leaf Disposal
 - a. Specification:
 - i. The machine shall autonomously pick up leaves and dispose of them in its own storage unit.
 - b. Rationale:
 - i. This eliminates the need for the user to repeatedly lift and carry weight, reducing the risk of neck, shoulder, and back injuries.
3. Battery Life
 - a. Specification:
 - i. The machine's battery shall last 120 minutes and recharge in 50 minutes
 - b. Rationale:
 - i. This ensures the battery lasts a 60-minute cleaning cycle, giving the machine extra buffer time in case it is a larger job or the conditions are unfavorable.²

¹ DeBruin, I. (2025, August 23). *How Long Does a Fall Yard Clean-Up Take?* - LambtonLawn LTD Blogs. LambtonLawn LTD Blogs. <https://lambtonlawn.com/blog/how-long-does-a-fall-yard-clean-up-take/>

²

<https://www.walmart.com/ip/Greenworks-80V-580-Cfm-at-145-Mph-Brushless-Leaf-Blower-with-2-5-Ah-Battery-Charger>

4. Battery Replacement

a. Specification:

- i. The machine shall feature a swappable battery that can be replaced in under five minutes.

b. Rationale:

- i. Allows users to resume machine operation after a quick battery replacement. The swappable battery system keeps the machine working even if another battery from the last use is charging.

5. Storage

a. Specification:

- i. The machine shall have sufficient storage to accommodate at least 30 gallons of leaves with a weight capacity of 50 lbs.

b. Rationale:

- i. This storage capacity aligns closely with the capacity of a standard leaf bag, reducing the number of times the leaves need to be disposed of. This allows the machine to work more efficiently with minimum interruptions.

6. Compaction

a. Specification:

- i. The machine shall compact collected leaves by a ratio of 1.5:1 in volume

b. Rationale:

- i. This increases the machine's storage capacity, reducing the frequency with which leaves will have to be disposed of, and allows the machine to operate with minimal interruptions.

7. Battery Per Runtime

a. Specification:

- i. The machine shall operate over a distance of 7,500 square feet in one charge, as according to conducted research, that is the average lawn size for suburban homes in the U.S.³

b. Rationale:

- i. Improves reliability and eliminates mid-cycle cleaning, charging, or battery replacement.

³ Wasson, S. (2022, August 23). *The Average Yard Size by State and City*. Today's Homeowner. <https://todayshomeowner.com/lawn-garden/guides/average-yard-size/>

8. Target Cost

a. Specification:

- i. The machine shall not exceed the sale cost of \$550

b. Rationale:

- i. Places the product competitively among existing products like leaf blowers and mowers, making it a financially viable option for homeowners. According to a survey, consumers are willing to spend an average of \$25 per cleaning. Assuming three cleanings per season and a machine life of eight seasons, the estimated value of the product is \$600, and \$550 is well below that.

If there was an alternative investment for removing up leaves, how much would you pay per cleaning?

125 responses

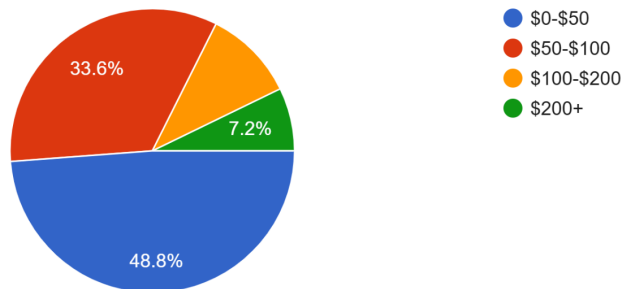


Figure D.10: Survey data for question “If there was an alternative investment for removing leaves, how much would you pay per cleaning”

Figure D.10 shows that approximately half of the surveyors would spend \$0- \$50 per leaf cleaning on a solution, and the rest would be willing to pay more.

Approximately how many times per season do you remove leaves from your yard in the fall?

127 responses

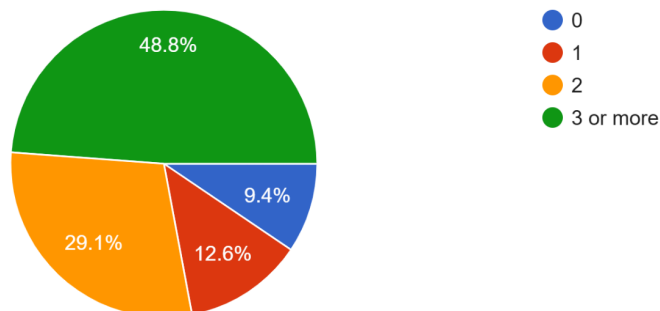


Figure D.11: Survey data for question “Approximately how many times per season do you remove leaves from your yard”

9. Noise Level

a. Specification:

- i. The machine shall not exceed 60 dB during operation, measured at a distance of 5 feet.

b. Rationale:

- i. This ensures the machine is suitable for residential neighborhoods, reducing the disturbance neighbors or users may experience during operation.

10. Durability

a. Specification:

- i. The machine shall be operational and efficient for at least eight fall seasons.

b. Rationale:

- i. This ensures long-term value for users, making the product economically sensible for them. This lifespan is also on par with other lawncare devices, with most typically lasting 8-10 years⁴.

11. Ergonomics

a. Specification:

- i. Any human interference shall minimize awkward lifting positions and not exceed 20 pounds.

b. Rationale:

- i. This reduces the user's risk of injury, addressing the primary issue. This also makes it more convenient for the user to use the product.

12. Autonomous Operation

a. Specification:

- i. The machine shall operate fully autonomously once it is set up and trained

b. Rationale: Automation is critical to minimizing physical intervention, reducing the risk of injury, and strain.

13. Operating Environment

a. Specification:

- i. The machine shall operate effectively outdoors in temperatures ranging from 30 degrees Fahrenheit to 75 degrees Fahrenheit.

b. Rationale:

⁴Minnesota Equipment. (2023, July 12). *How Old Is Too Old for a Lawn Mower?* - Minnesota Equipment. Minnesota Equipment.

- i. This reflects typical fall weather conditions in the United States⁵, ensuring that the machine's performance during peak season is efficient and satisfactory.

14. Natural Conditions

a. Specification:

- i. The machine shall operate with resistance to conditions like dust, moisture, and uneven terrain

b. Rationale:

- i. This is necessary for outdoor operation, preventing wear and machine issues due to natural conditions. If not accounted for, the machine can face mechanical and electronic damage, leading to system crashes⁶.

15. Safety Precautions

a. Specification:

- i. The machine shall have an emergency stop in the event of malfunctions, debris clogging, or unexpected terrain.

b. Rationale:

- i. This provides immediate machine shutdown in dangerous situations, protecting users and their property from potential harm the machine can cause.

16. Maintenance

a. Specification:

- i. The machine's maintenance shall be limited to battery replacement and minimal inspection once per fall season.

b. Rationale:

- i. Limiting maintenance to only battery replacement and minor inspections reduces user responsibility and encourages continued use.

⁵ *Fall Temperature Averages for Each USA State - Current Results*. (n.d.). [Www.currentresults.com. https://www.currentresults.com/Weather/US/average-state-temperatures-in-fall.php](https://www.currentresults.com/Weather/US/average-state-temperatures-in-fall.php)

⁶ Hulley, L. (2024). *The Impact of Dust in Electronics*. [Bulgin.com. https://blog.bulgin.com/blog/the-impact-of-dust-in-electronics](https://blog.bulgin.com/blog/the-impact-of-dust-in-electronics)

Appendix B: Brainstorming Ideas

Before a final design was created, many ideas were brainstormed and sketched. 45 were created in total, but they were filtered to the designs below to determine the final design.

Brainstorming Ideas

Below are simple drawings that were created of the ideas brainstormed. Each drawing is simple, furthering the process of creating a final design.

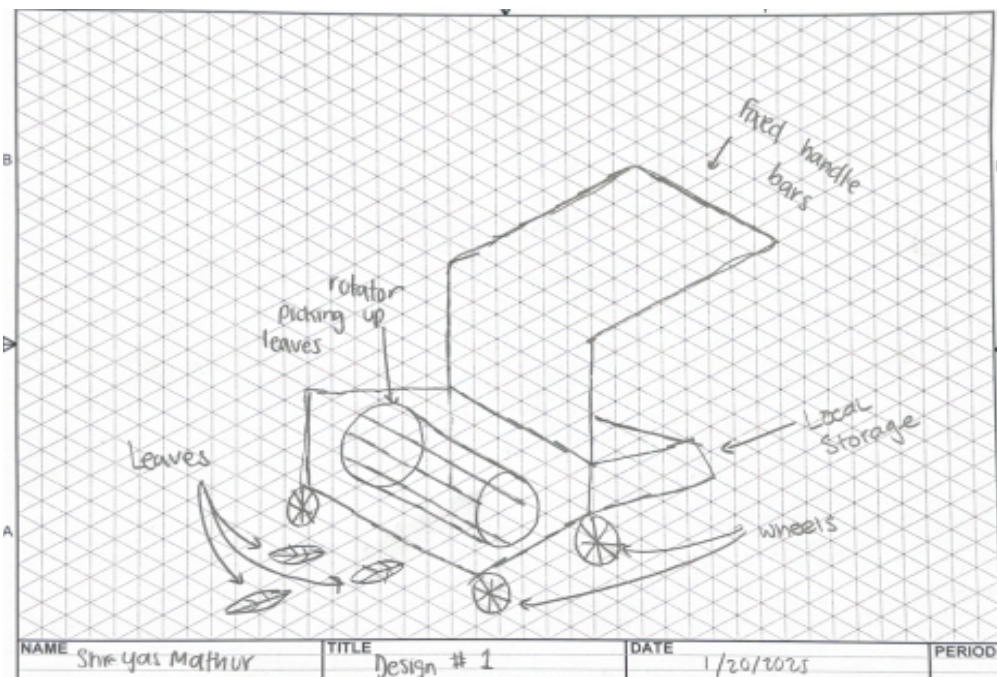
Sketches	Description
 NAME Shreyas Mathur TITLE Design # 1 DATE 1/20/2025 PERIOD	- The machine is manually pushed - Leaves are collected at the bottom - Leaves are stored in local storage and will have to be dumped into a bag - Handlebars, intake, and body material are plastic

Figure D.12: Leaf cart

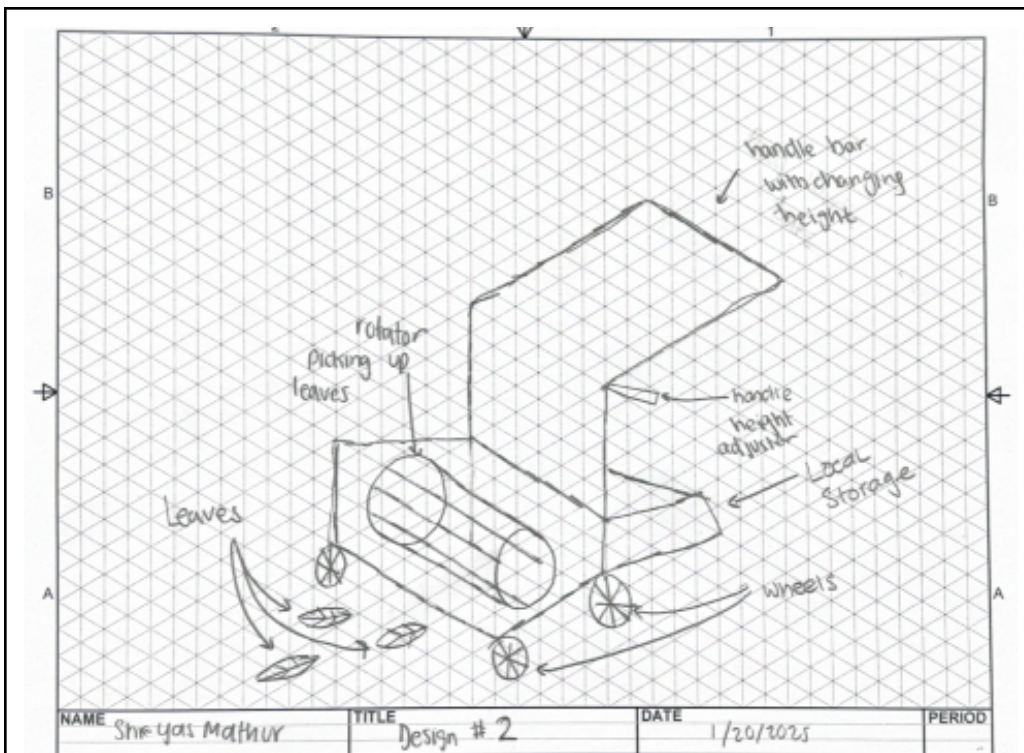


Figure D.13: Leaf cart modified

- Figure D.13, except that there are adjustable handlebars to improve posture for the user

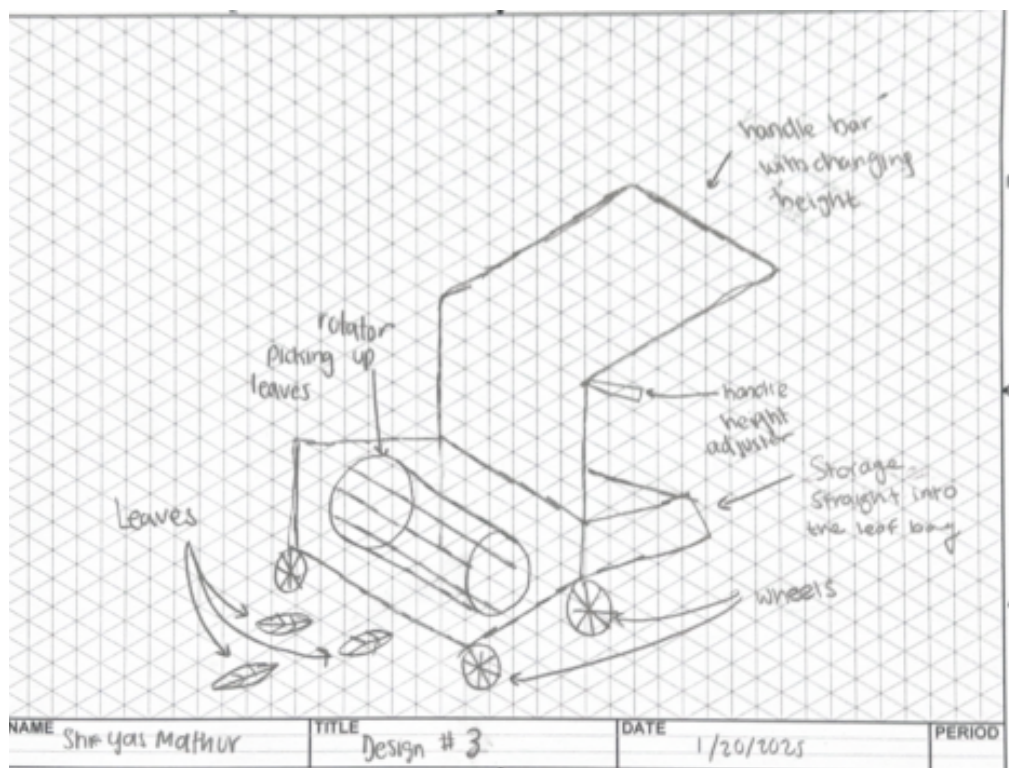
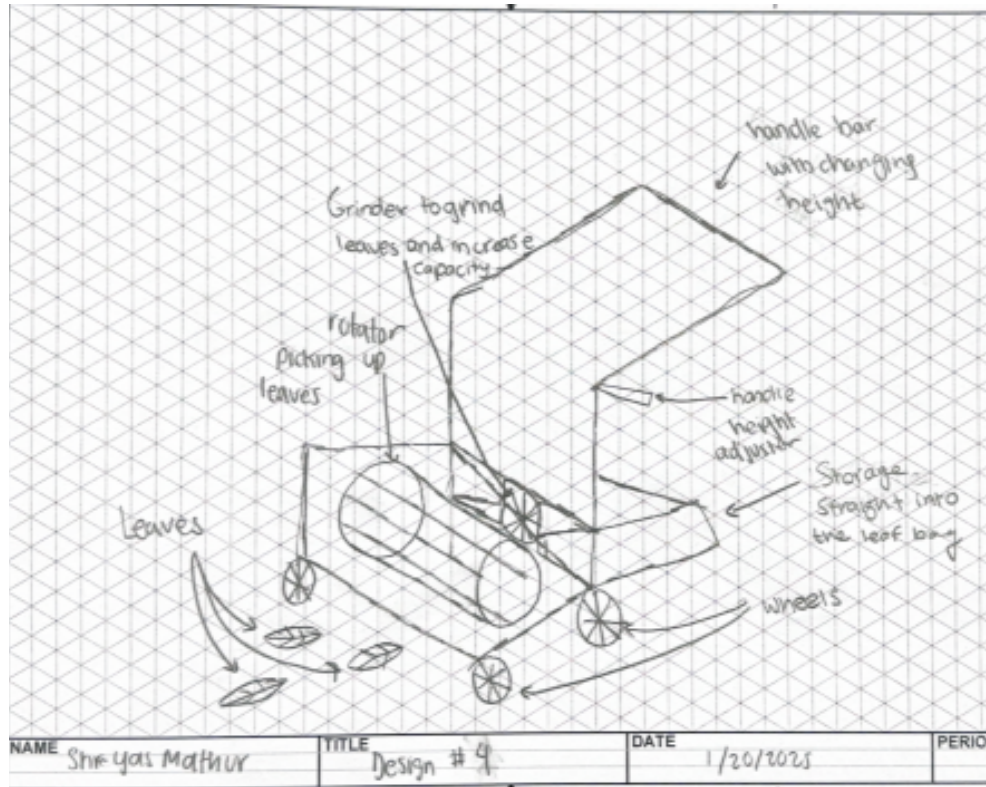


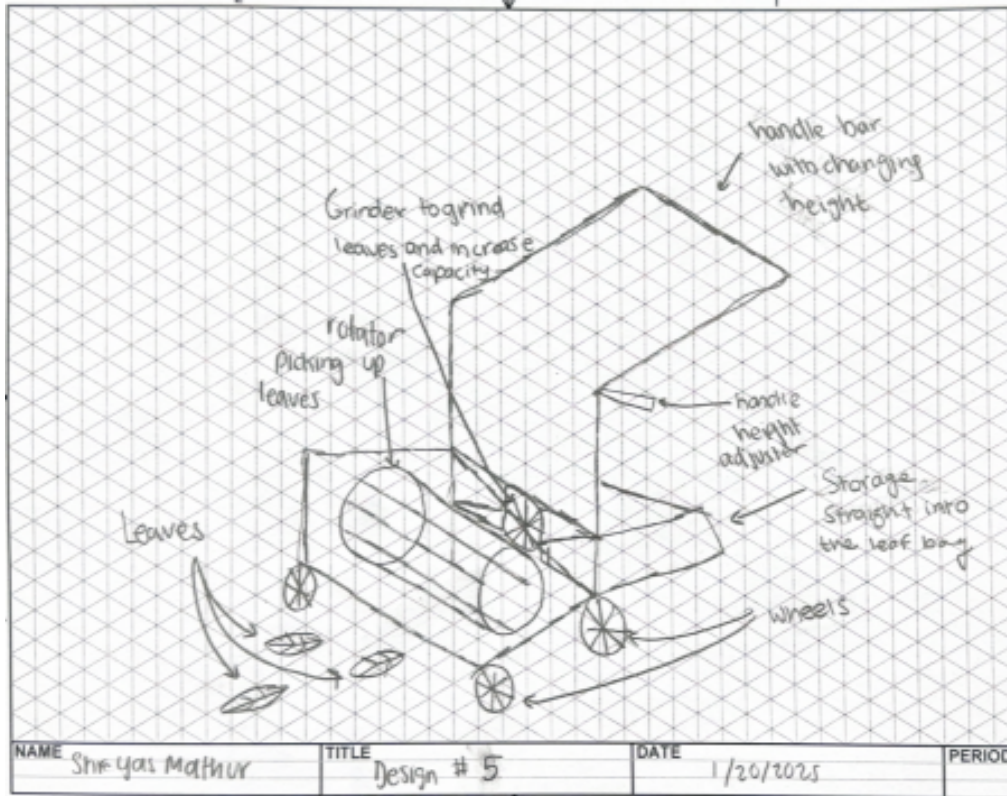
Figure D.14: Leaf cart modified

- Figure D. 14, but leaves are stored straight into leaf bags instead of local storage



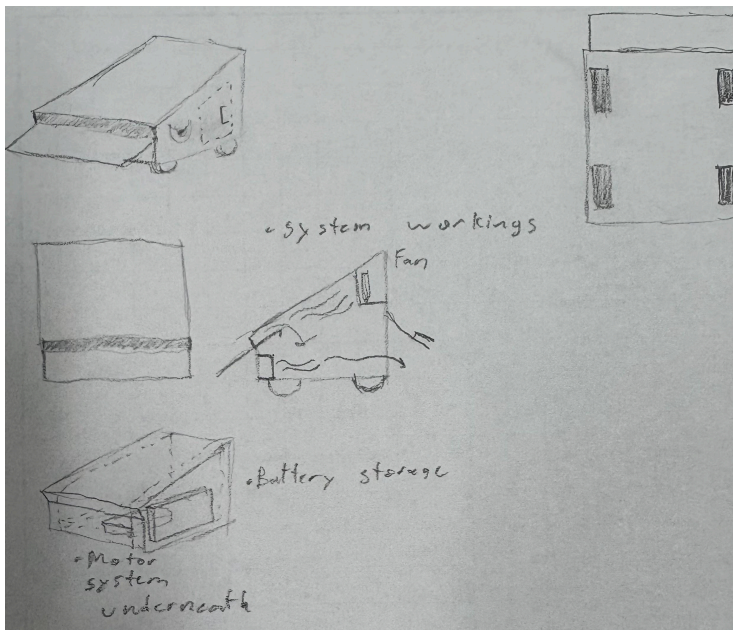
- Figure D.15, but there is a grinder to cut leaves and reduce the empty space

Figure D.15: Leaf cart modified



- Figure D.16, but the intake material is changed from plastic to rubber

Figure D.16: Leaf cart modified



- Figure D.17 is an autonomous vacuum and storage unit
- Motor drives throughout the lawn, letting the incline pick up the leaves into its storage unit
- Back door compartment to automatically dump leaves

Figure D.17: Incline leaf removal robot

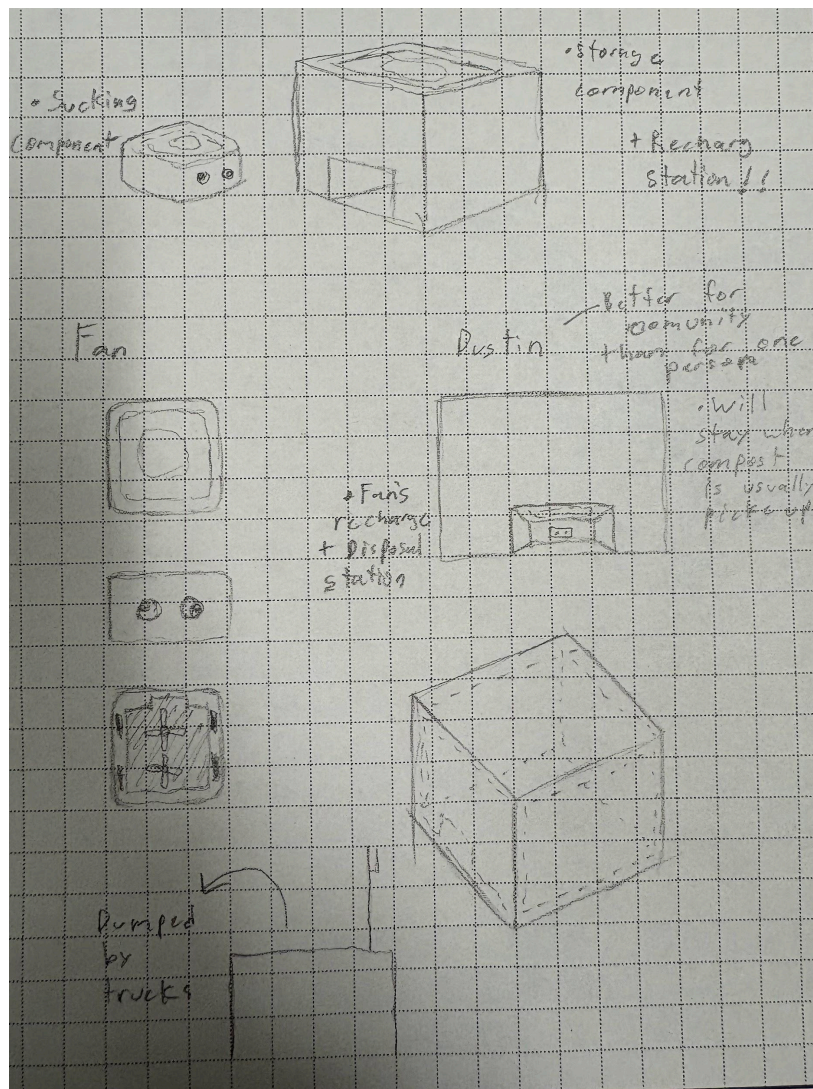


Figure D.18: Two system leaf vacuum and storage

- Figure D.18 is two separate machines: the smaller design to remove and dispose of leaves, and the. The larger component is to be the storage unit and charging station
- The smaller component is small. storage, needing back and forth trips to the larger storage unit
- Larger component is the main storage for leaves while also being the charging station for the smaller component

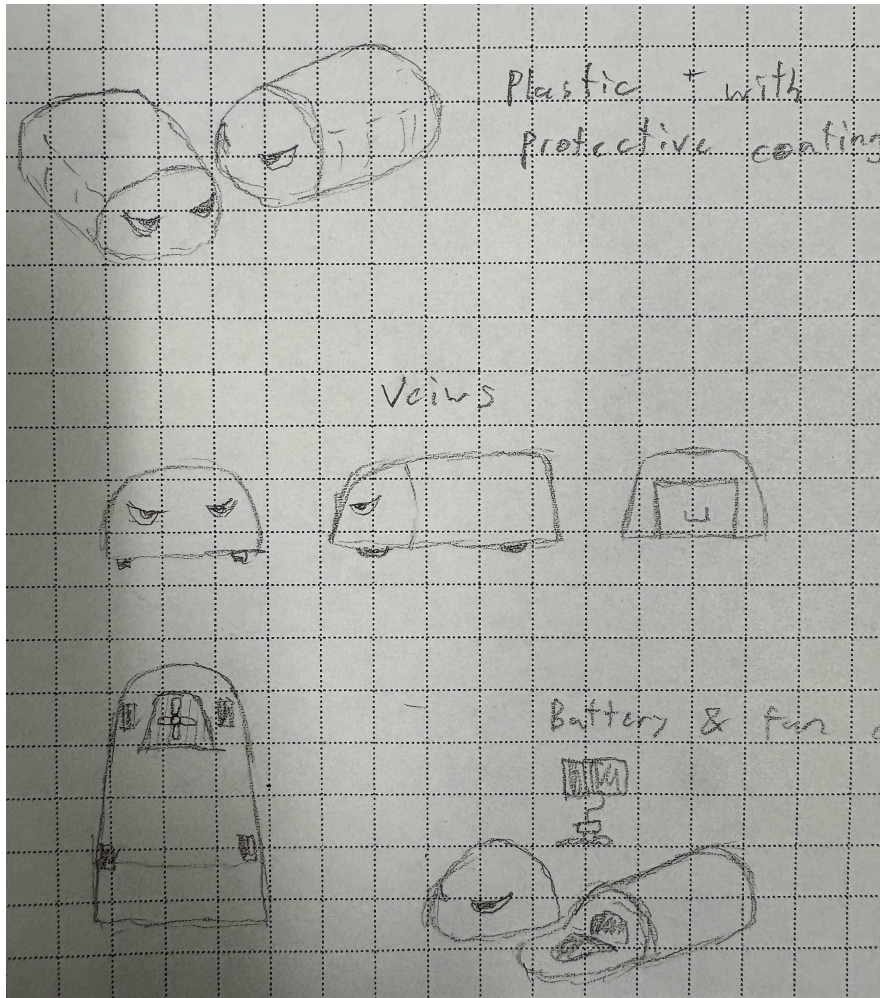


Figure D.19: Mini autonomous leaf vacuums

- Figure D.19 shows two identical autonomous robots that remove and disperse machines
- Instead of one large machine, two smaller designs were created, allowing more area of coverage on a lawn
- Holds a vacuum at the front, storage unit in its back and baback-and-forthpose leaves in the designated area

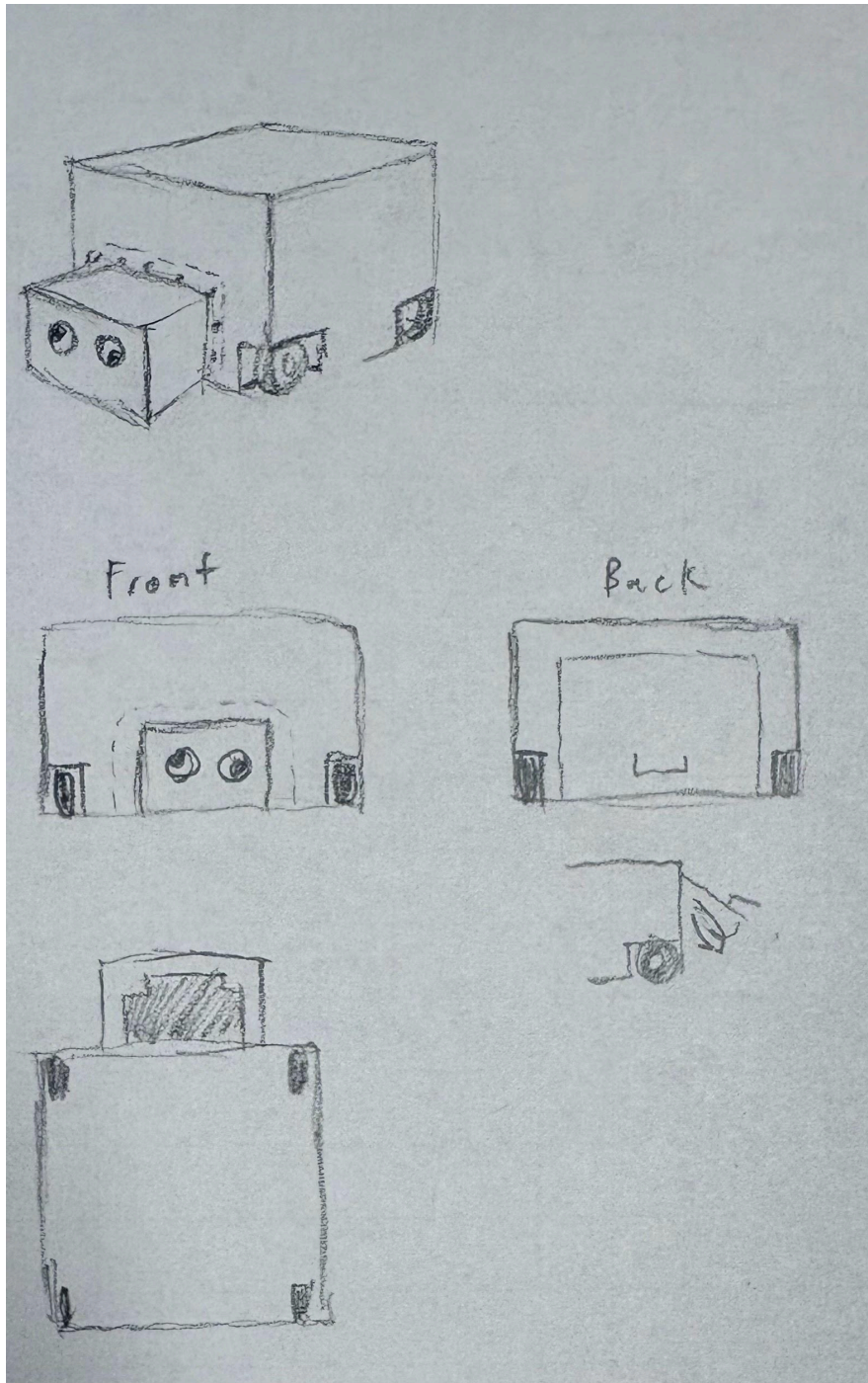
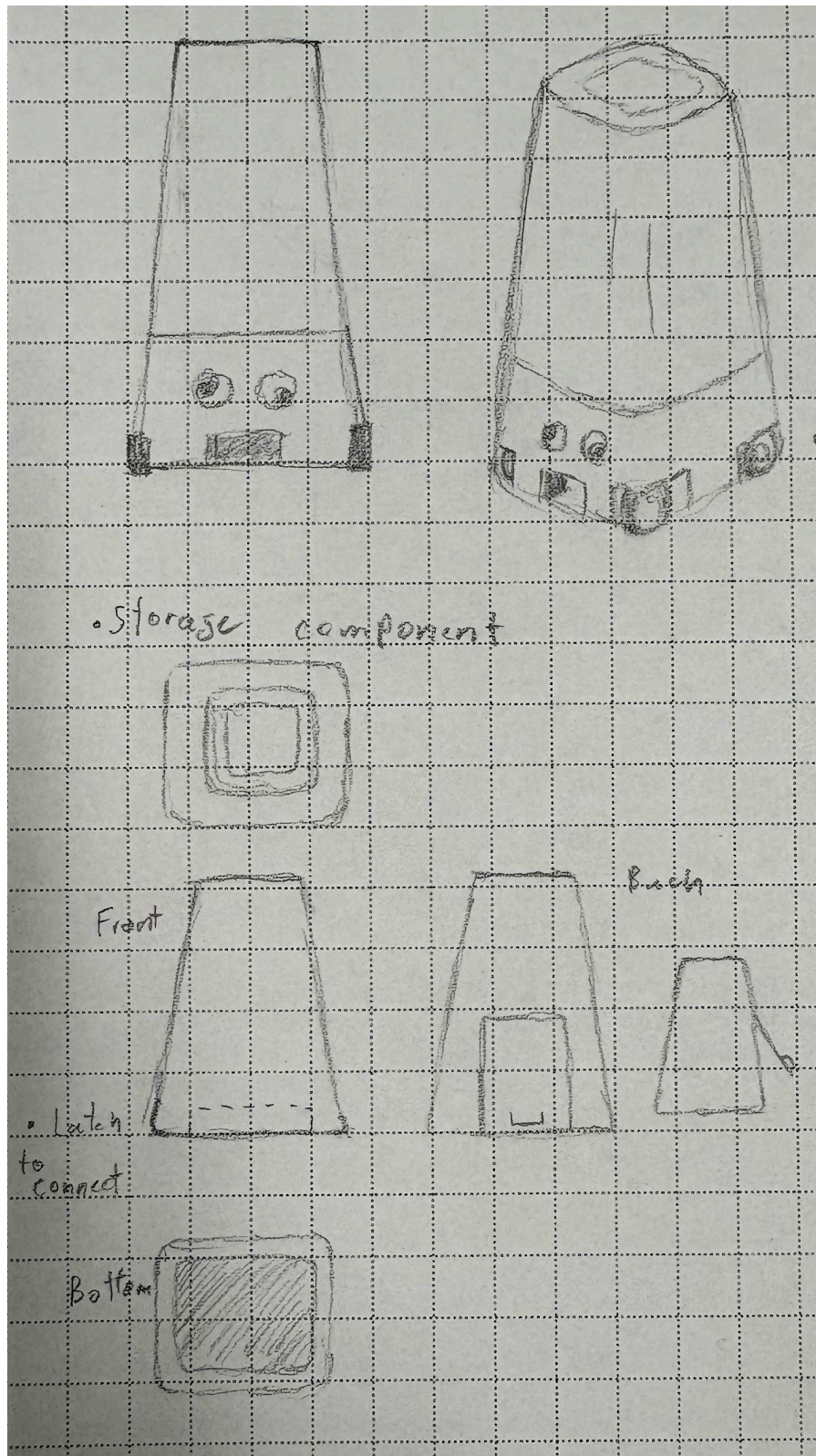


Figure D.20: Tank leaf vacuum and storage

- Figure D.20 is a large, industrial design that holds both a vacuum and a larger storage unit in one component
- An autonomous design that can finish a large lawn without having to dispose of leaves as often
- Requires more battery life than other designs



- Figure D.21 is an autonomous leaf vacuum with a large storage unit.
- Two part machine. The lower half is the motor and vacuum.
- The entirety of the top of the machine is for storage.
- Back panel to easily remove leaves into a bag, side of the road, etc.

Figure D.21: Autonomous leaf vacuum and storage

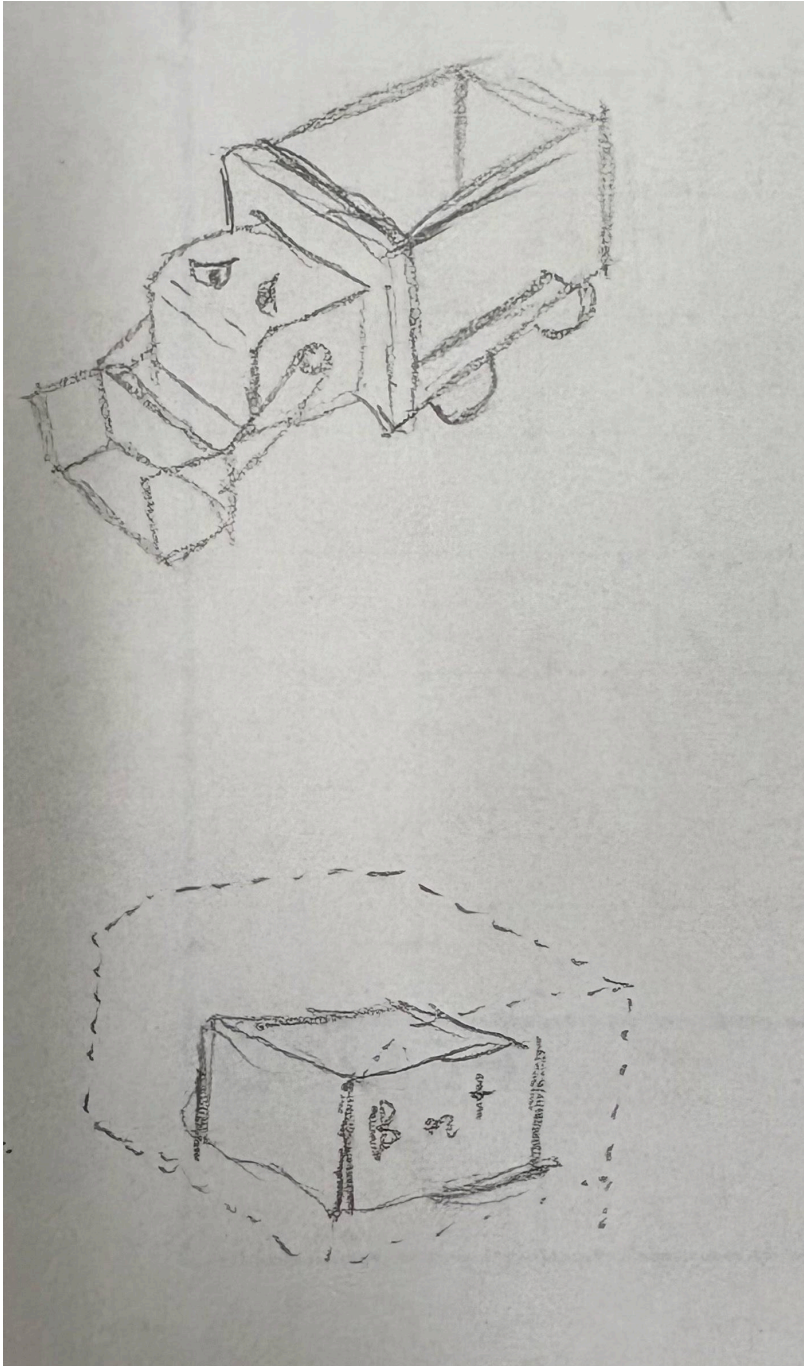


Figure D.22: Leaf garbage truck

- Figure D.22 is an autonomous robot that has an open storage compartment and a scoop mechanism for the leaves to be picked up by
- Holds a larger battery unit, allowing longer life for figure D.#
- Open storage unit, once full, can lift and dump leaves into a designated area

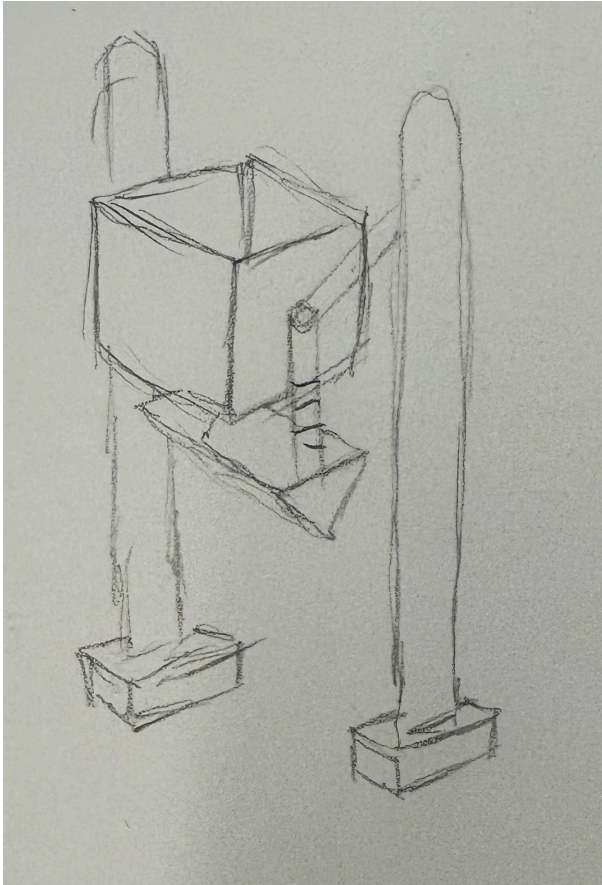


Figure D.23: Leaf pendulum

- Figure D.23 is a manual design that uses weight to move and remove leaves
- The bottom compartment swings down to pick up a group of leaves, and when it swings up, it will dump the leaves into the central storage unit, then swing down and repeat the process
- The motion will also push the legs forward, allowing figure D.# to move manually

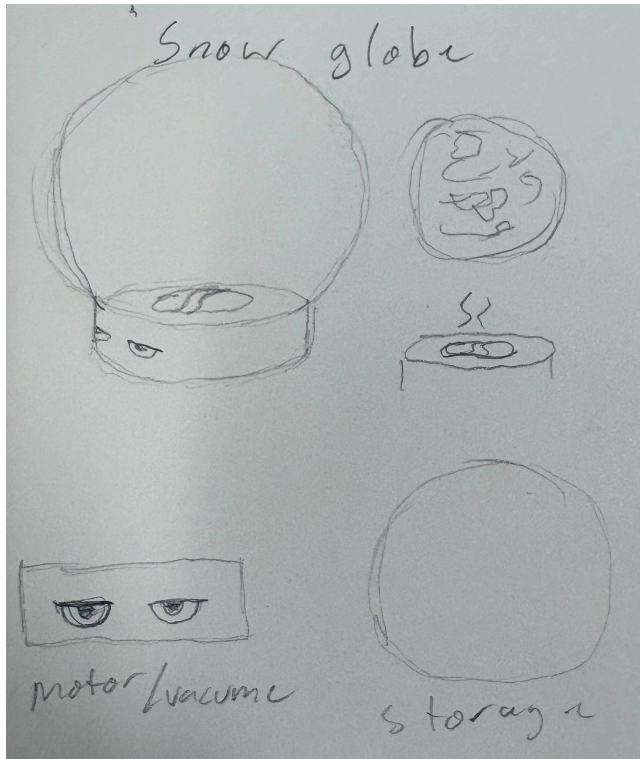


Figure D.24: Snow globe leaf vacuum

- Figure D.24 is an autonomous robot that, on the bottom, is the vacuum and motor unit, and above is its storage unit for leaves
- The cylindrical component moves along the lawn with a vacuum in the center of its diameter
- The large, spherical components will be the storage unit for the leaves, having a large amount of open space for them

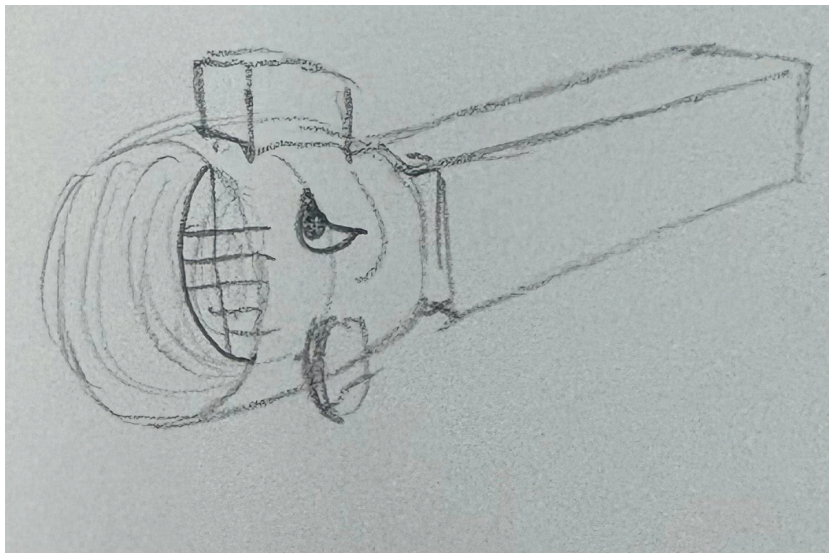
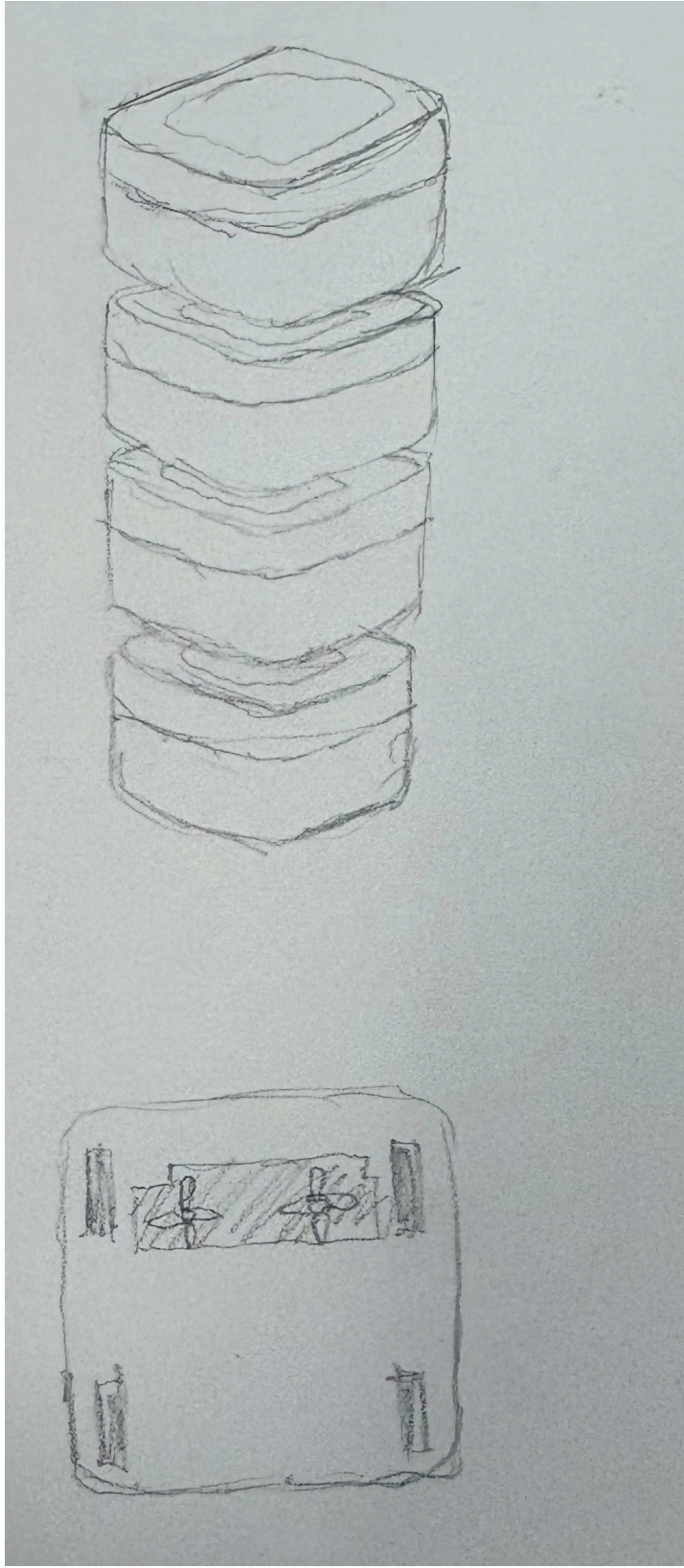


Figure D.25: Mulcher autonomous robot

- Figure D.25 main components are the cylindrical machine at the front
- The battery sits on top of the component, while in the center is a mulching component
- Behind the component is a reusable leaf bag, acting as the storage unit for the leaves



- Figure D.26 are four, identical autonomous machines that work together to remove leaves in a lawn
- Each machine can be stacked on one another in a charging station
- Uses teamwork to remove all the leaves in a yard
- Vacuum and storage unit inside the machine, and on top is where leaves are disposed from

Figure D. 26: Leaf vacuum robots

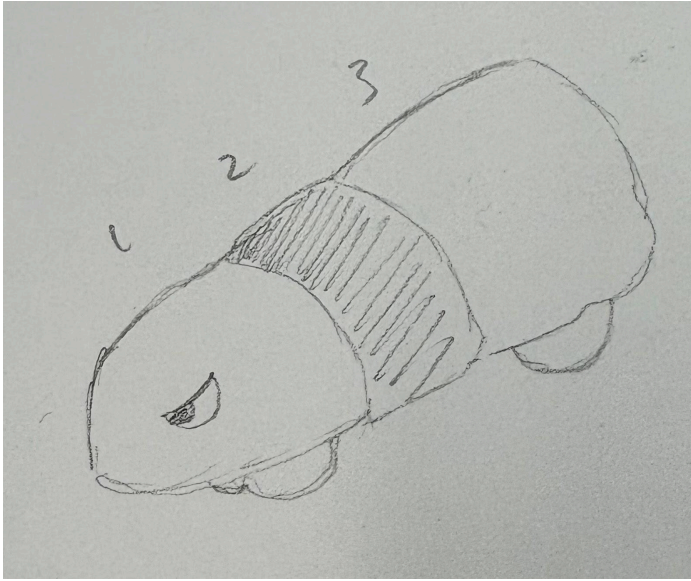


Figure D.27: Autonomous leaf vacuum and storage

- Figure D.27 is a large, three-sectioned autonomous machine
- The front section is the vacuum, motor, and battery
- The middle section is for mulching
- The back section is for the storage and disposal of leaves

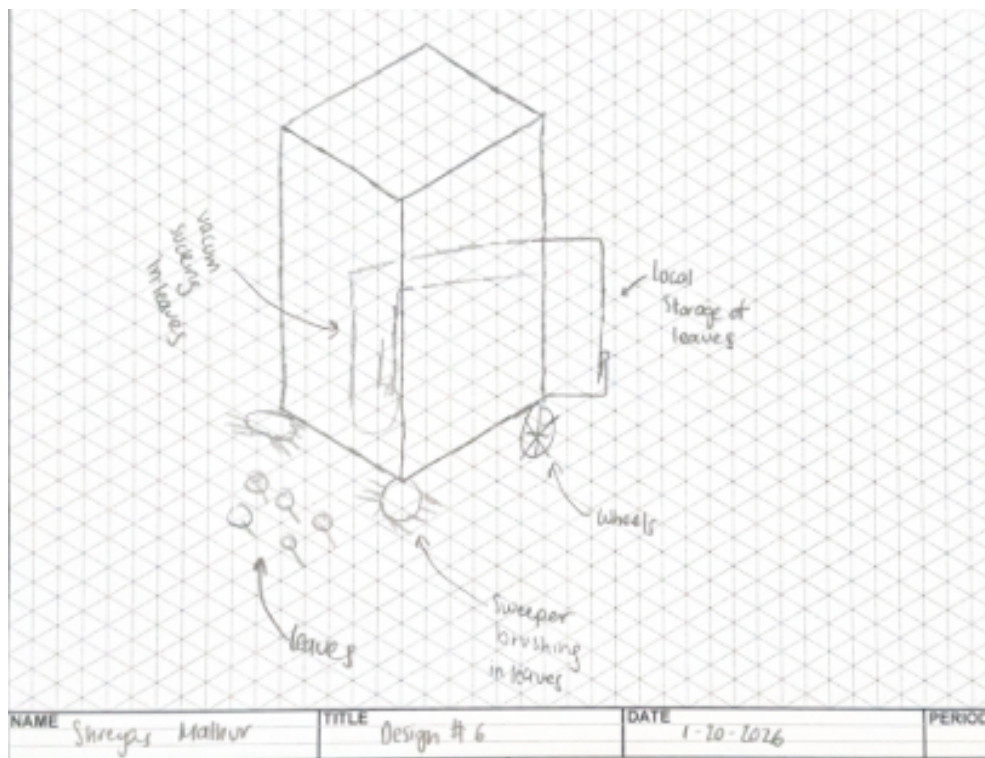
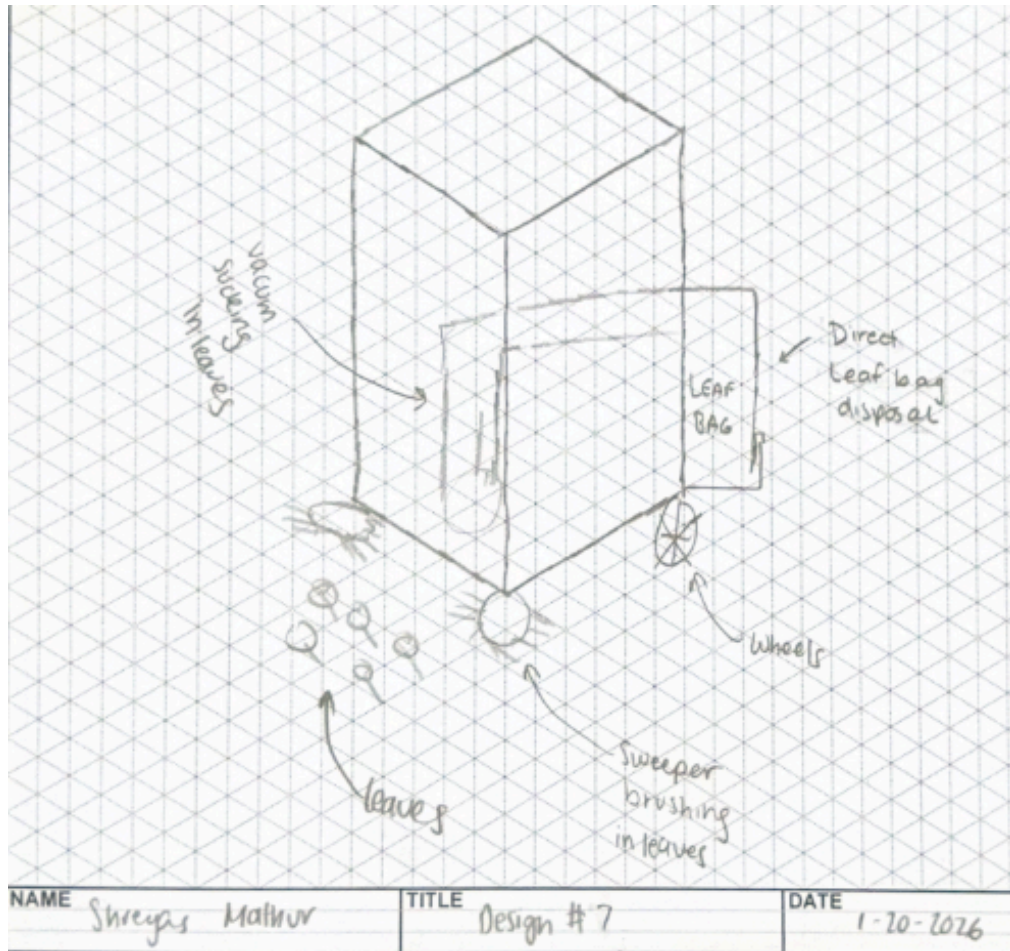


Figure D.28: Auto Fridge

- The machine is autonomous
- The sweeper directs leaves towards the machine
- Vacuum sucks up leaves and stores them in the local container
- Leaves must be manually dumped into the bag



- Figure D.28, except the leaves are stored directly into a leaf bag instead of a local container

Figure D.29: Auto Fridge modified

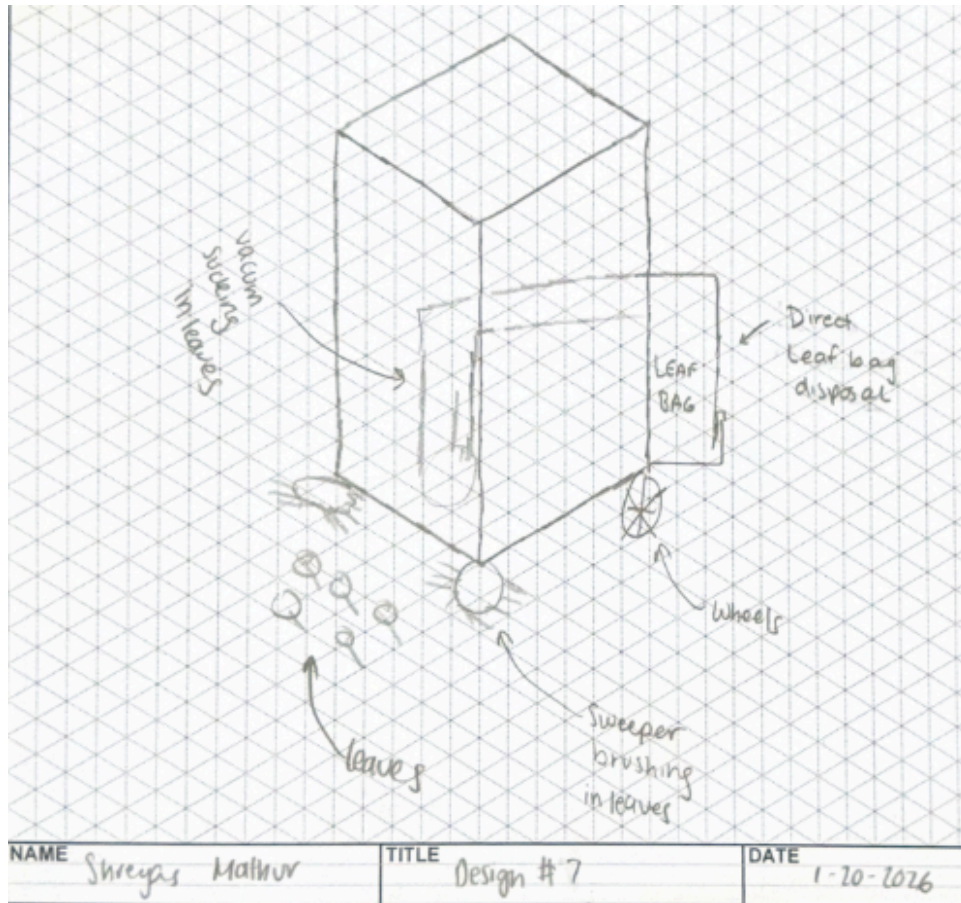
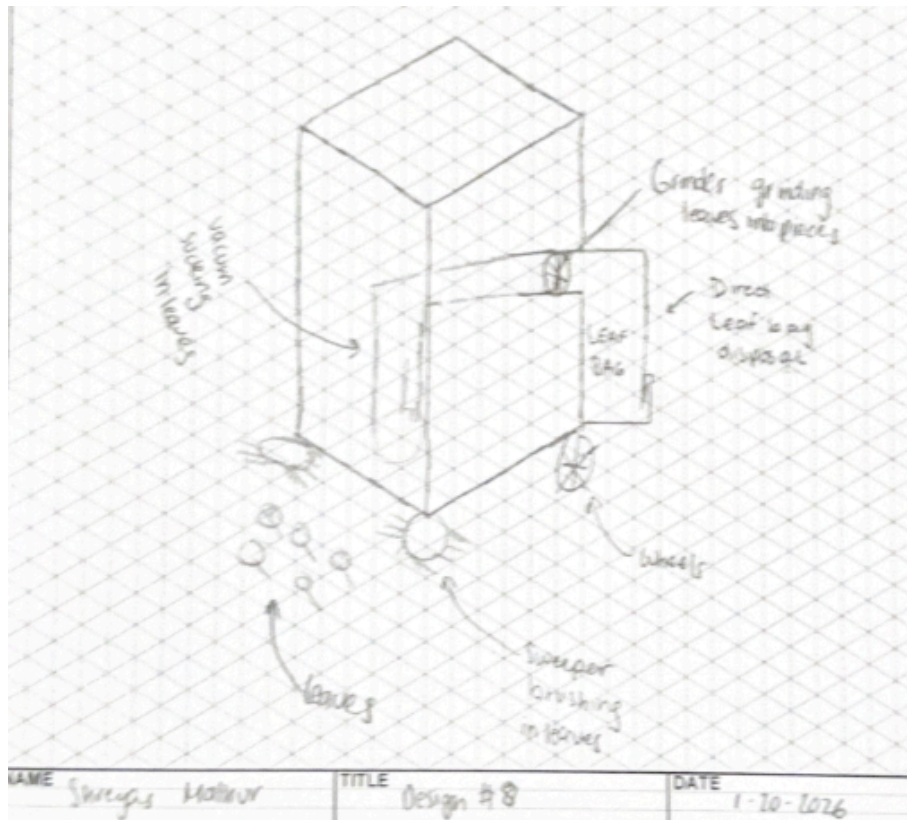


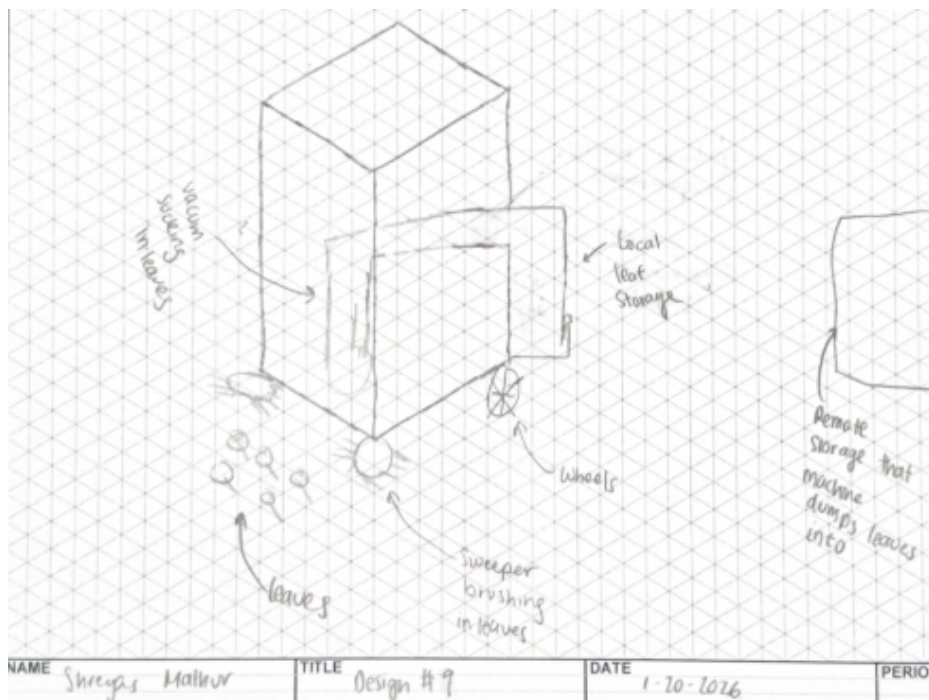
Figure D.30: Auto fridge modified

- Figure D.29, except that the leaves are stored directly into a leaf bag instead of a local container



- Figure D.30, except that there is a grinder to grind the leaves before they reach the storage area

Figure D.31: Auto fridge modified



- Figure D.31, except the machine goes and dumps leaves in a remote storage container after local storage is full. Leaves must be manually moved from this to a bag

Figure D. 32: Auto fridge modified

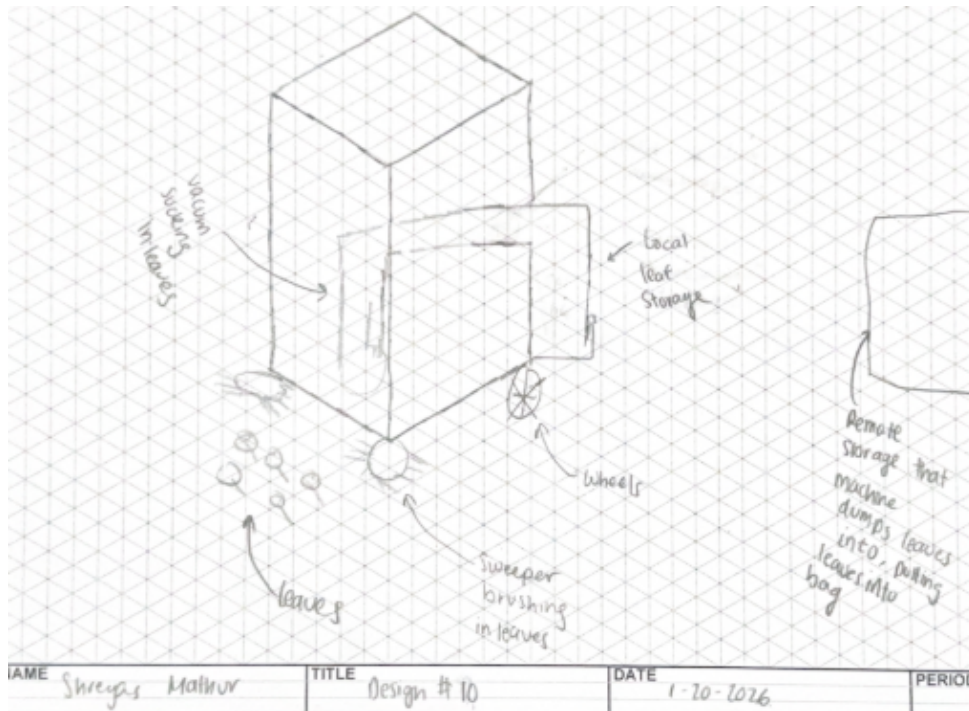


Figure D.33: Auto fridge modified

- Figure D.32, except that the remote storage puts leaves into a leaf bag by itself.

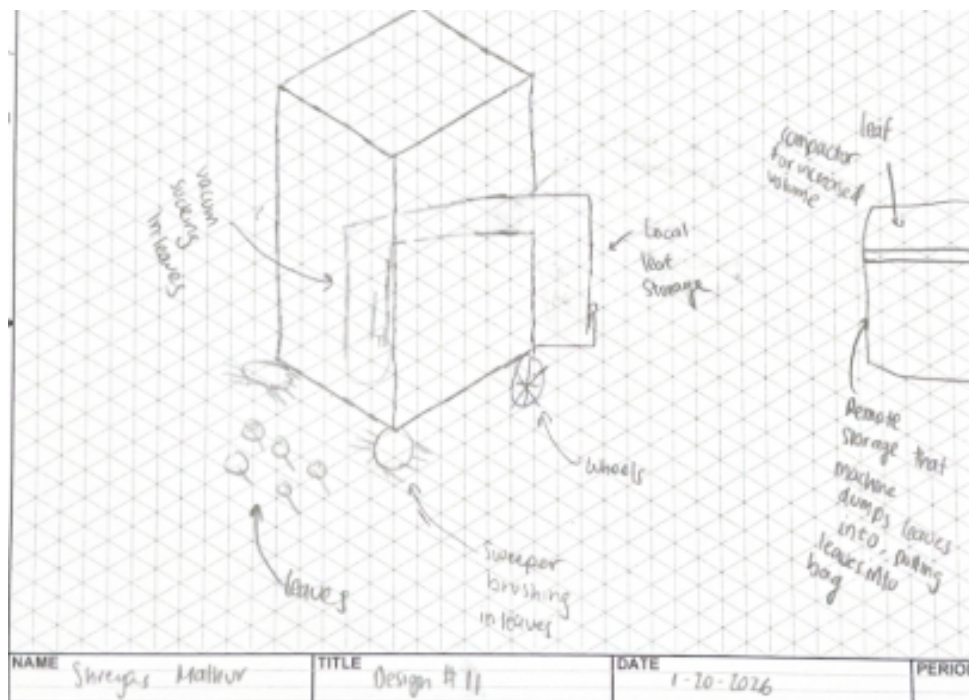


Figure D.34: Auto fridge modified

- Figure D.33, except that the remote storage has a compressor to compress leaves down to decrease volume

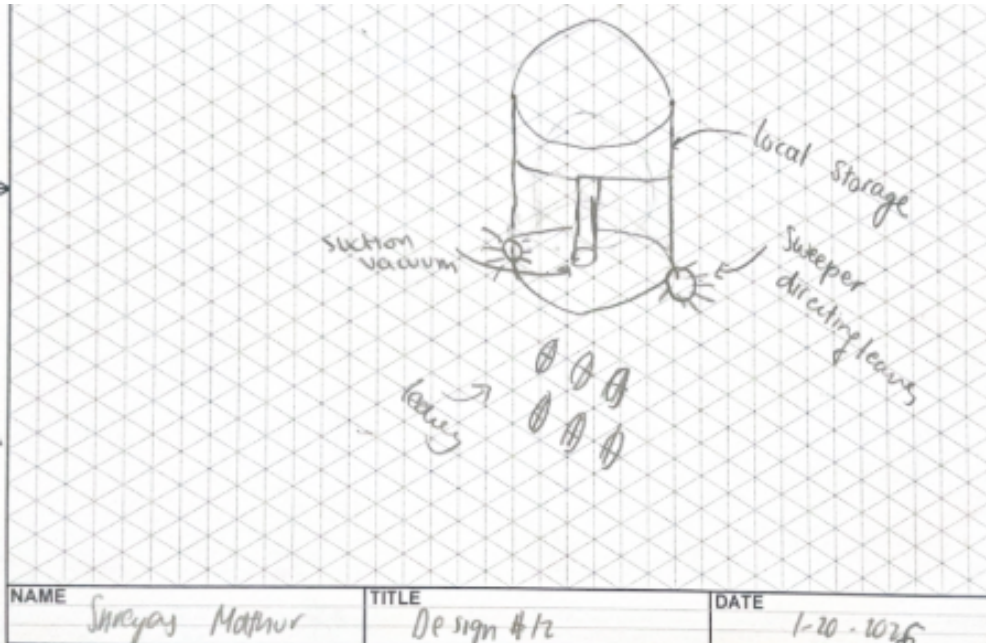


Figure D.35

- Figure D. , except the scale is largely reduced to be more compact and space efficient

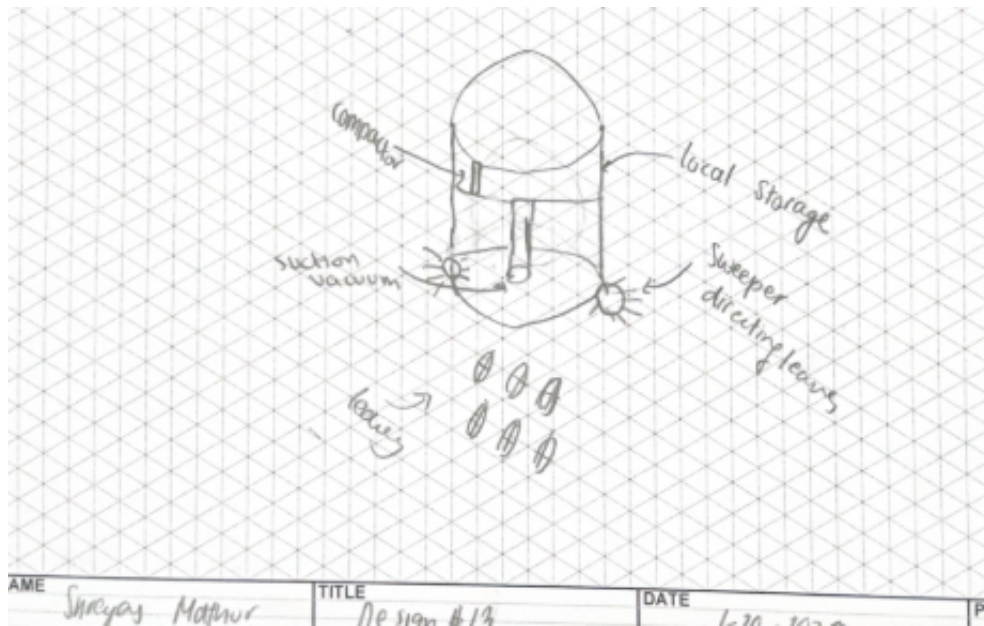


Figure D.36

- Figure D.28, except there is a compactor to reduce leaf volume

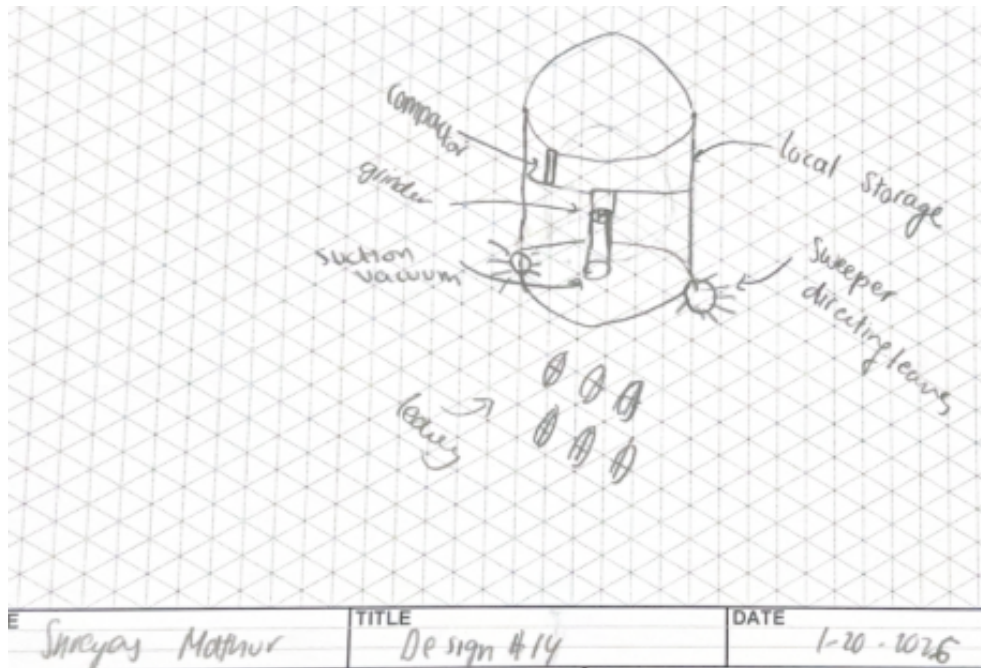


Figure D.37

- Figure D.36, except there is a grinder for the leaves before it gets to storage

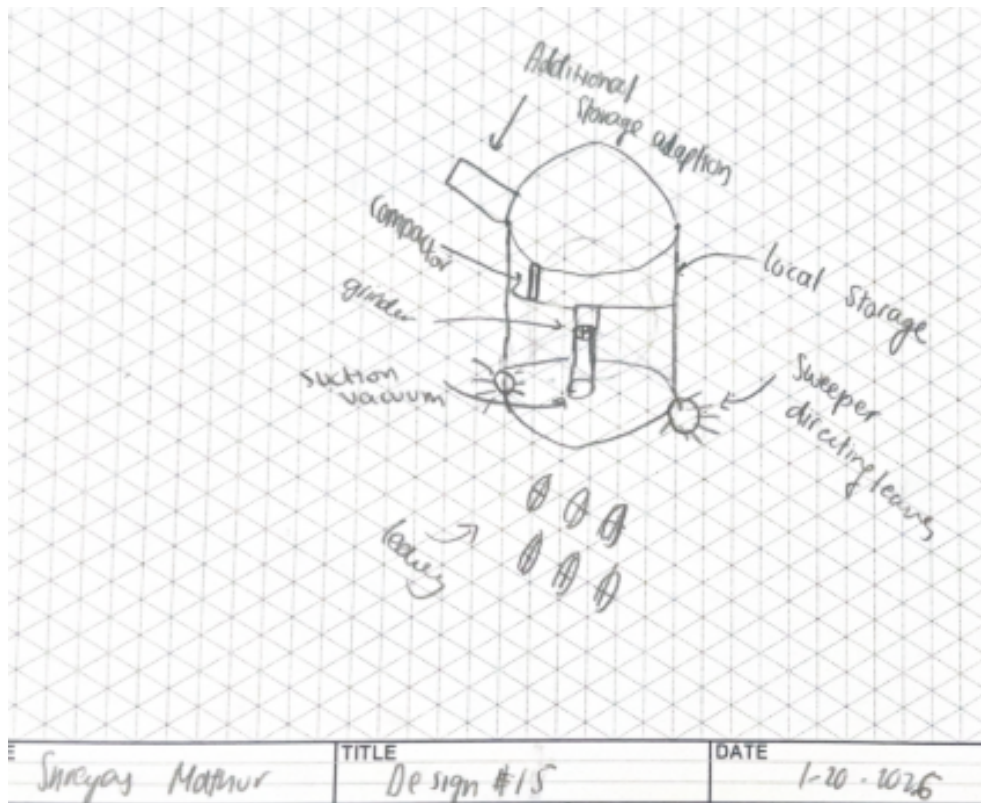


Figure D.38

- Figure D.37, except there is an attachment to add additional storage bags onto the machine to increase capacity.

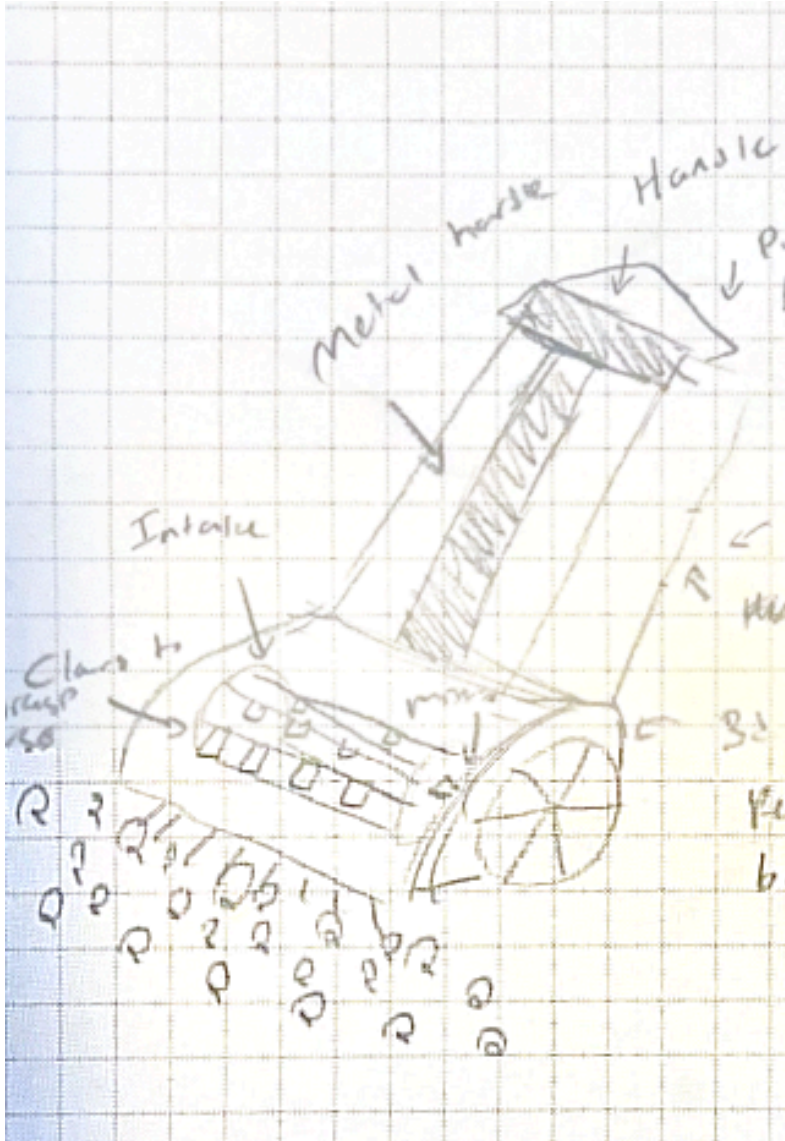


Figure D.39: Lawn sweeper modified

- Figure D.39, but changing the bag into a heavy-duty felt option
- Figure D.39, but adding a flap to the bag for easy disposal
- Figure D.39, but changing the material of the handle from metal to hard plastic for weight
- Figure D.39, but changing the motor of the intake to work via mechanical power and not battery
- Figure D.39, but changing the color of the whole thing to a more earth tone
- Figure D.39, but changing the material of the blades that suck the leaves from metal to soft plastic
- Figure D.39, but adding a shredder or compressor mechanism that allows more leaves to be collected

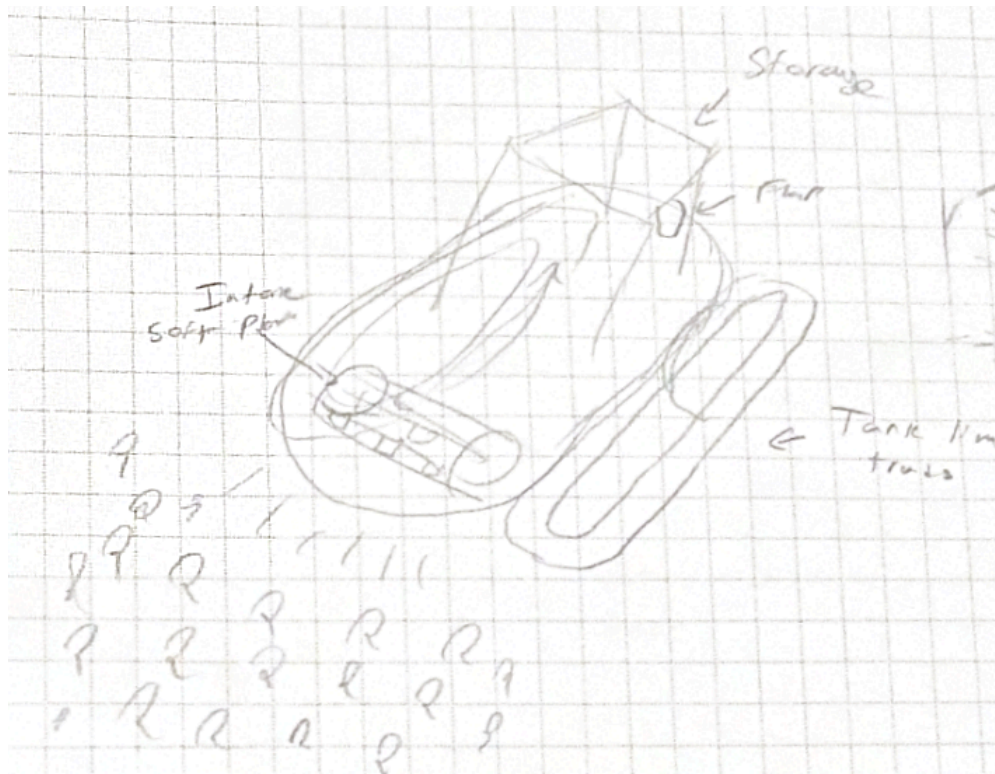


Figure D.40: Lawn vacuum

- Figure D.40, but adding a charging station that can be placed outside so the user doesn't have to charge it on their own
- Figure D.40, but make it so the intake does not have to use a separate motor from the wheels or compressor
- Figure D.40, but make the components more modular so they can be cleaned more easily
- Figure D.40, but changing more of the internal components to plastic to save weight
- Figure D.40, but changing the color of the whole thing to black
- Figure D.40, but changing the treads on the wheels to be more heavy-duty because of the elements
- Figure D.40, but adding a shredder to take up less space
- Figure D.40, but changing the storage system to a different side, or put on the back or inside if it can be shredded enough

Appendix C. Product Concept Designs

Product Concept #1: The Pusher

Designer: Seth Jones

Product Description:

A self-autonomous, 3x2x1-foot, robot. This product will be equipped with a 3x.5x3 inch metal panel at its front to push leaves onto the side of the road. It is for users who have the ability to have leaf curbside pickup in their neighborhoods. The body of the product is a half-cylindrical metal frame (radius 6 inches) that holds its power source and motor, as well as the coding inside it. It will have four plastic wheels attached on each corner on the inside of the machine. On the front of the body is a hook, allowing it to connect to the metal panel (and disconnect for storage purposes). On the back of the body is an on and off switch. Please turn on when the rods are turned on, so Tayniche can pair and know where its borders are. Small metal rods (x10) will be placed across the border of the lawn to signify a boundary for the product. There will also be a pair of “goal point” rods. Goal point rods will be placed along the edge of the lawn, leading to the road. This lets the product know where to push all the leaves to be picked up. It will be able to push leaves of a 2,500 sq ft lawn to the side of the road, allowing the neighborhood's leaf removal to pick the leaves up and dispose of them.

Operation:

Setting up markers across the user's lawn, as well as a set of goal markers to indicate where to push the leaves to. Once the rods are set up and turned on, proceed to turn on Tayniche 1 while also having the metal panel attached. Set the product down, and let the process begin. Once finished, proceed to flip the switch on the product off, and collect the rods. Store however you need.

Inputs (Rods):

- Two Triple A batteries
- Plastic covering
- Physical planting of rods across lawn
- Manual activation (on and off button)

Internal Process (Rods):

- The on/off button, located on the top of the rod, will be pressed, sending a signal to the other rods
- Identifies and creates a border, messaging the product to stop and turn around once it meets the border
- Goal rods will be indicated with a white strip of tape. Goal rods will be placed on both ends of where the user would like to leave them (on the edge of the lawn, leading to the road).

- The product will be sent a signal of where the goal rods are, given its destination to push the leaves to

Outputs (Rods):

- Signal creates a border to the product, preventing it to leave the lawn entirely
- Energy from the two Triple A batteries are taken
- Manually remove rods when the process is complete and turn off
- When batteries die, replace them

Inputs (Product #1):

- Four Triple A batteries inside the shell of the product
- Bluetooth module 100W signal device to connect to rods
- A brushless motor is installed for the wheels

Internal Process (Product #1)

- The on/off switch will be flipped once all rods are placed and turned on
- A signal is sent from rods to the product, allowing the process to begin
- The brushless motor, using the energy from the batteries, moves forward
- The path the product will take is pre-coded into the product, only difference will be the shape of the lawn

Outputs (Product #1):

- Energy taken from the four Triple A batteries
- A cleared-out, leafless yard
- Slight wear from the removal process

Justification:

This product was chosen as one of the final solutions to pursue because of how feasible it is to build while still following most of our design requirements.

The product will be autonomous, preventing injuries from happening and saving more time for the user to spend their day as they would like. But, the feasibility to build this self-autonomous robot was greater than the other designs. Other designs were too complicated to be built, as well as saving money within the budget given. Tayniche 1 saves time for the user and time to be built, as well as money to be built and purchased, unlike the other designs. These two factors affect both groups: the engineers and the users. For the engineers, it is a more realistic design that could be built. Fewer parts needed to buy, simpler code for instructions, and easier to test. For the users, it only takes minimal setup, saving more time as if the user were to remove their leaves like before. The product will cost less to build, leading to a lower cost to purchase on the market compared to other designers, if they were selected to move forward.

A constraint for this product is that it can only be used by users who have curb-side leaf removal. Tayniche 1 still takes over the need for human interaction to push leaves onto the side of the road.

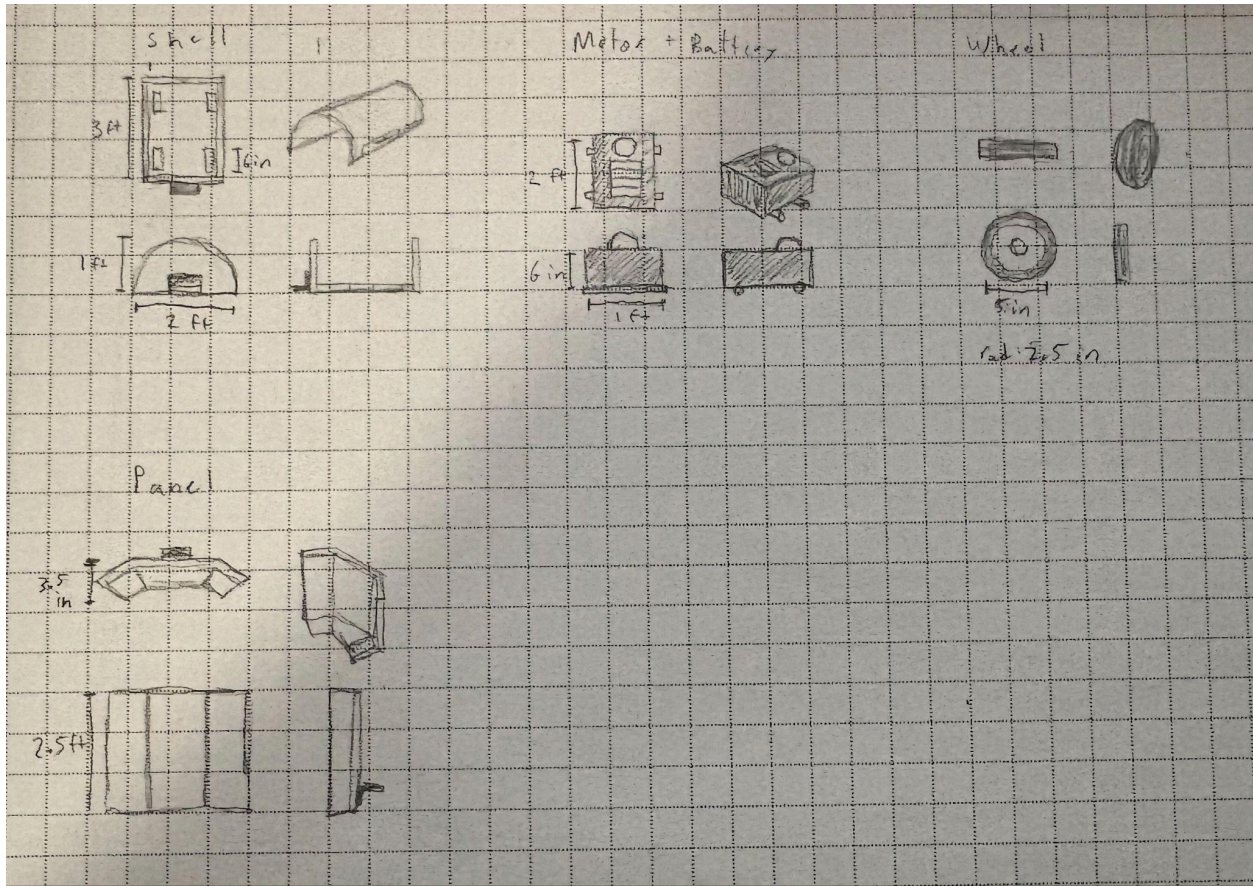


Figure D.41: Automatic leaf pusher design

Product Concept #2: The Gulper

Designer: Seth Jones

Product Description:

This product is a manual, 2x2x2-foot leaf removal device. It is a box hollowed out through the center, leaving a thickness of 1.5 inches, and an incline of 1 foot, leading to the inside of the box. The one side with the incline will also be the side with the handle, meant to let leaves slide in. On the other open side of the box is a reusable 6x2x2 foot bag, made out of textile material. The box will be made out of refurbished plastic, and two rubber wheels will be attached on both sides of the box. A plastic handle that can extend to 4 feet will be attached on the box, allowing the user to pull it along.

Operation:

The box will be rolled along the user's lawn, being pulled by the handle. The incline inside the box will let the leaves slide in and enter the bag attached to the other end of the box. Continue pulling until the bag is full. Once the bag is full, remove the bag, attached by hooks on the box, and dump the leaves into a yard waste bin (or wherever else the user disposes leaves). Reattach the bag, and continue the process until leave removal is completed.

Inputs:

- Human interaction
- Reusable bag

Internal Process:

- Hold onto the handle and pull
- Wheels rotate, making it easier for the user to pull
- Incline picks up leaves
- Pull forces push leaves into the bag

Outputs:

- Full leaf bag
- Human interaction to dispose of leaves within leaf bag

Justification:

This product was chosen as one of the final designs because it saves money, decreases the time to remove leaves, prevents injury, and is feasible to build.

The product saves money in both cost compared to other products, and the budget to build the product. It is not autonomous, giving more focus on how to reduce the physical strain of removing leaves appears less. One constraint, however, is that it requires human interaction, going against one of the design requirements. This product was still chosen because it is something that can be built.

Feasibility was another factor, because a self-driving automobile design is a stretch for the budget. Feasibility is also a factor towards money, letting more money in the budget to be saved for building this product.

Preventing injury is another need that this product has. It requires a pull motion to operate, and the strength to lift and dump a 6x2x2-foot bag. Less bending down movements is required to remove leaves, and all the leaves will already be stored in a bag.

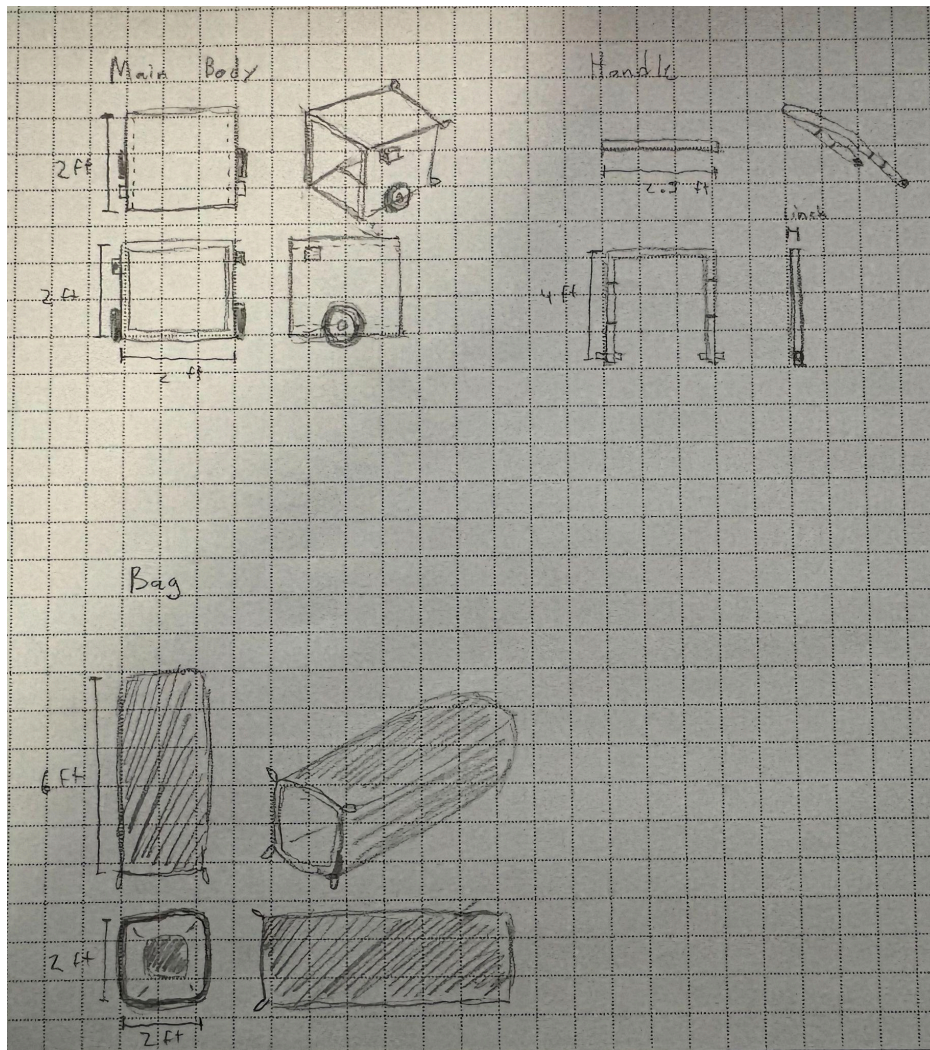


Figure D.42: Leaf gulper concept design

Product Concept #3: Auto Fridge

Designer: Shreyas Mathur

Product Description:

This product is an autonomous electronic leaf-picking machine designed to collect leaves with minimal user effort. It uses an electronic vacuum system to suck leaves up from the ground and stores them in an attached leaf bag. The machine is battery-powered, and the battery is located at the top of the machine for quick replacement and lasts a full cleaning. The product works by following a path preset by the user on an app on their cell phone. The purpose of a replaceable battery is to make sure that the machine can continuously run, and even if one battery dies, another one can be placed in and keep continuous cleaning instead of stopping for recharging.

Operation:

The inputs for the product are electrical energy from the rechargeable battery and a map of the path from the user. Once the user sets the cleaning path and starts the machine, it will follow the given path and operate on its own, minimizing human effort as much as possible. Internally, the sensors work to navigate the given path and make sure that there are no obstacles in the way that can get stuck in the machine. As the machine moves, the vacuum sucks up leaves and transports them into a leaf bag, which is easily accessible for the user.

Justification:

The problem is that leaf collection methods like raking cause too much physical strain on the user and result in various back and neck injuries because these methods require repeated bending. This product directly works against the problem by performing this task independently, allowing users to set up the cleaning area using their phone and not be involved in any laborious activity.

This concept was chosen as one of the final solutions because it eliminates user interference and is the most convenient for them. Other solutions, like manual methods and leaf blowers, still require physical effort, putting the body in physical stress. This autonomous design, on the other hand, will eliminate all of these problems and shall not cause any injuries.

This satisfies the customers' needs by saving time, preventing injury, and reducing hassle. This saves time because the user does not have to continuously work on this task, giving them time to work on other things while this task is being done autonomously. It reduces hassle and prevents injury because the user is not doing any of the work themselves, reducing the opportunities for injury.

This design meets all of the design specifications. However, many specifications, like noise, can only be confirmed after the product is tested.

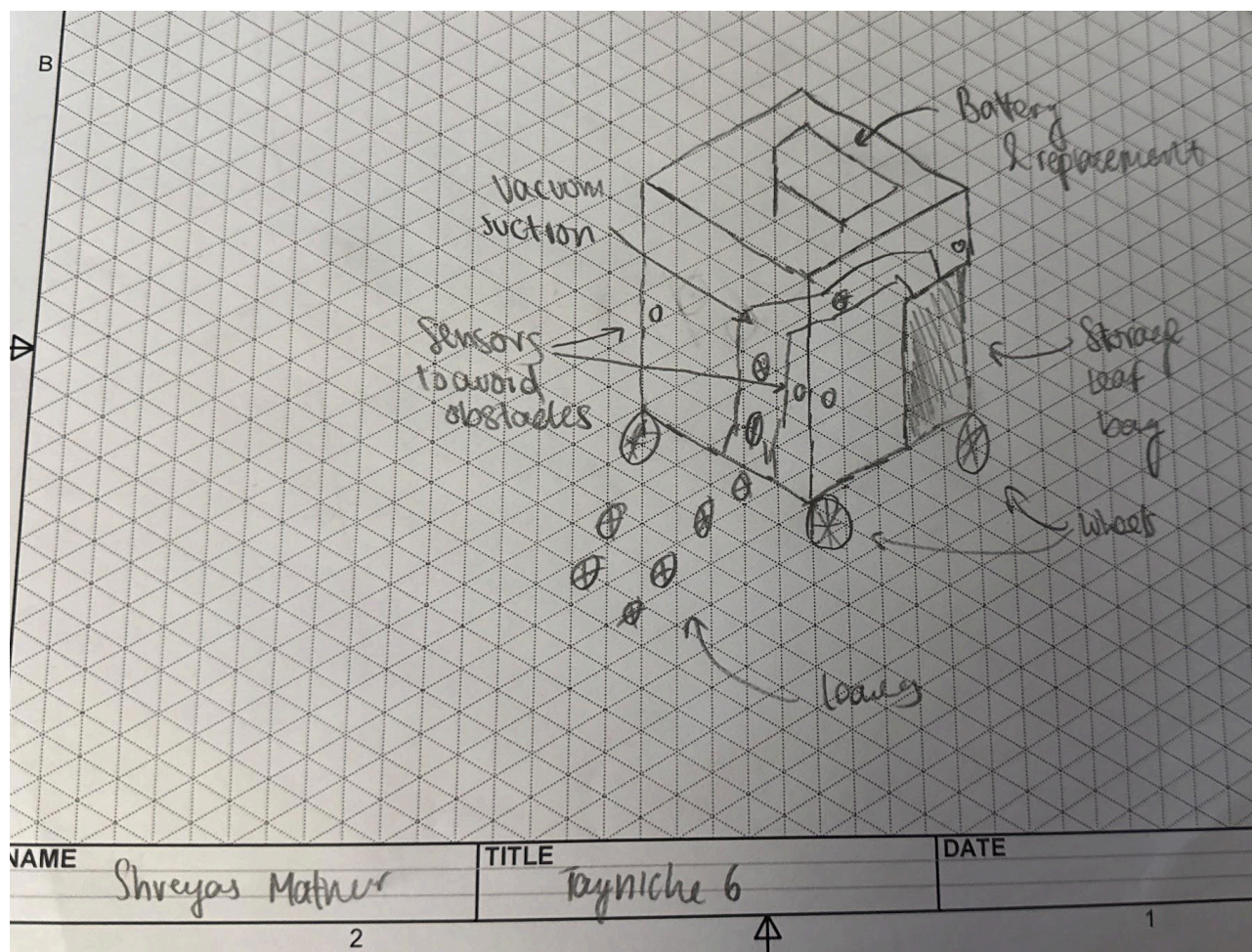


Figure D.43: Auto Fridge Concept Sketch

Product Concept #4: Cart Pusher

Designer: Shreyas Mathur

Product Description:

The product is a manual leaf-collecting machine that is designed to pick up leaves efficiently. The primary function of the machine is to pick up leaves through its rotating mechanism, which lifts the leaves and stores them in a leaf bag attached to it as the user pushes it. The machine also has an adjustable handle, allowing users of different heights to operate it with comfort and proper posture, reducing physical strain on the user. The final appearance of the product is going to be compact and sturdy, with a metal frame, and it does not take up much space in the user's storage.

Operation:

The input is human force. When the user pushes the machine forward using the adjustable handle, it provides the machine with mechanical power to spin the rotator at the base. Then the rotator comes in contact with the leaves, it lifts them up, and the internal mechanisms guide the leaves into a leaf bag for storage. The leaf bag is attached using clips and ensures easy replacement after it is full to minimize physical strain.

Justification:

The problem is that leaf collection methods like raking cause too much physical strain on the user and result in various back and neck injuries because these methods require repeated bending. This product directly works against the problem by having a mechanism that picks up the leaves using mechanical power and minimizes human effort. The handle that the user has to push is adjustable, allowing users to adjust it to their desired setting to ensure an upright position when they clean up their lawn.

This design was chosen because it has the perfect balance between effectiveness, simplicity, and usability. This machine is very effective and does not require electricity or fuel to operate, which reduces the complexity of the product and ensures easier maintenance as well. The usability of this product is also better than current solutions because it keeps the user in a comfortable position the entire time that they are working, preventing injury and reducing tiredness at the end. Many alternative solutions use electricity or fuel sources for power, which increases the complexity of the product and makes it harder for the user to use, as well as increasing noise and costs.

This design also meets many of the design specifications, only lacking in the component of it being autonomous, which would require complex electronics and sensors to be included, which increases maintenance and chances of the machine breaking down.

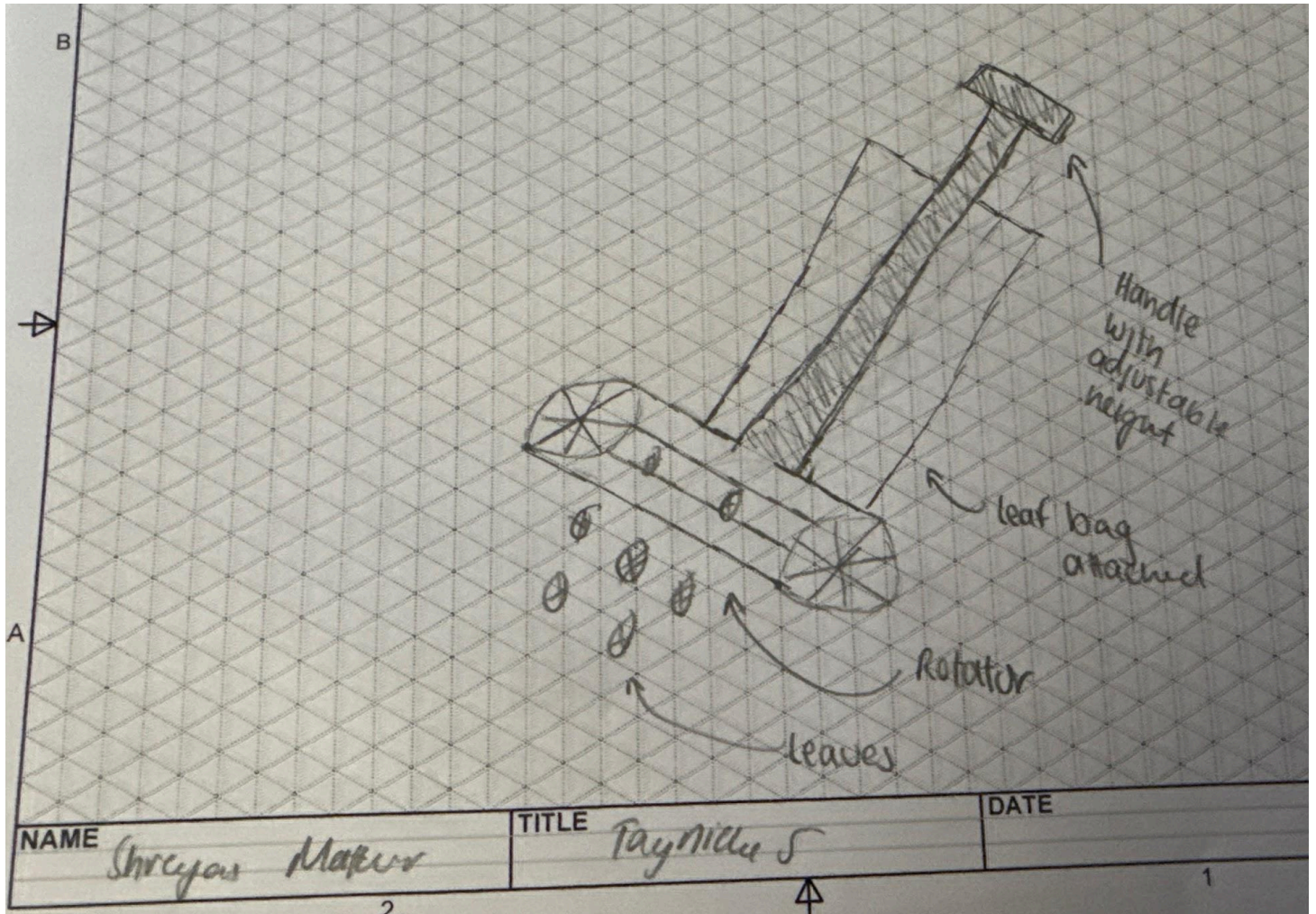


Figure D.44: Cart Pusher Concept Sketch

Product Concept #5: Lawn Sweeper

Designer: Tanish Lakshmanan

Product Description:

This product is a manual leaf collector that is pushed and designed to gather leaves without relying on much manual labour. The device has a rotating intake with soft plastic blades that pick up leaves, which are pulled into a heavy-duty bag with a flap for easy disposal. The design is lightweight and durable overall.

Operation:

The product uses small motors to drive the mulching process, but overall, it relies on user push power, transferred from the wheels through gears to spin the intake blades. The user pushes the device to collect the leaves, then the device mulches the leaves and stores them in a bag.

Justification:

This design solves the problem of physically demanding and time-consuming leaf cleanup in yards. Compared to traditional methods like raking and bagging, this device saves time and reduces hassle. The use of mechanical power eliminates the need for heavy power consumption, increasing the product's reliability. The heavy-duty felt bags also increase overall durability, making this a strong solution for people who want a simple, reliable way to pick up leaves.

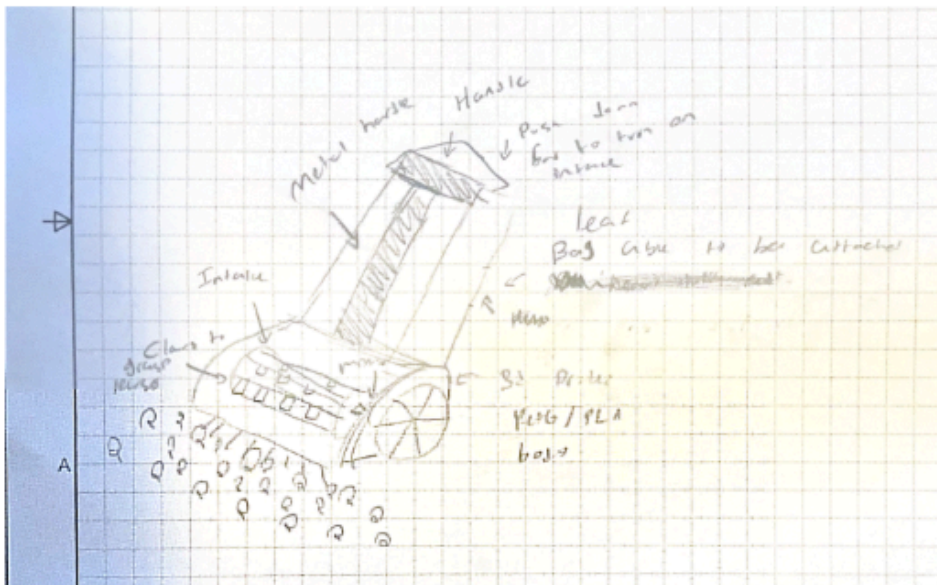


Figure D.45: Lawn Sweeper

Product Concept #6: Leaf Tank

Designer: Tanish Lakshmanan

Product Description:

This product is an autonomous leaf-collection device designed to collect leaves with minimal user involvement. The device uses powered tank tread wheels and another motor to spin a rotating intake that brings in the leaves, which then go through a mulching process that shreds them to take up less space. The device is charged by an outdoor charging station connected to an outdoor outlet.

Operation:

The device operates on electrical power supplied by a rechargeable battery charged by the outdoor charging station. The device wakes from rest in the charging station when the user wants it to start, then goes around the yard, detects leaves, picks them up, and stores them in a compartment above the device.

Justification:

This concept significantly improves convenience and efficiency in leaf collection. By being autonomous, the product saves time and reduces hassle for users who do not want to perform manual labor. The modular design makes cleaning easy. Shredding the leaves increases carrying capacity, which leads to less emptying.

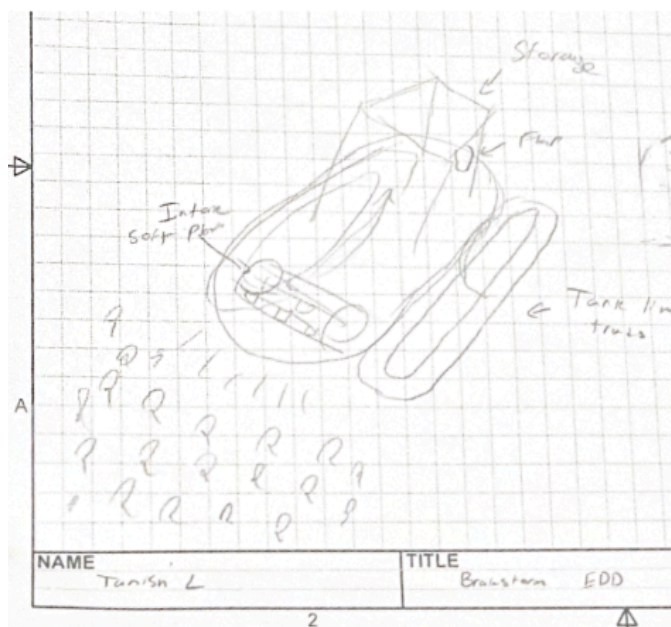


Figure D.46: Autonomous leaf vacuum

Appendix D. Decision Matrices

To determine a final design and solution, a decision matrix was necessary in the selection process. The figure below shows the designs described in Appendix C, and each was tallied where the highest score per designer was selected to be in the survey from Appendix D. Each product was scored from one to five in each category, one being the lowest/worst score and five being the highest/best score.

Criteria	Gulper	Pusher	Auto Fridge	Cart	Lawn Sweeper	Leaf Tank
Cost	4	1	2	3	4	1
Ease of Use	3	4	4	3	3	4
Buildable	5	2	2	4	5	2
Effectiveness	4	3	3	3	4	3
Differ from Market	4	3	4	2	2	3
Storage In Garage	5	3	2	1	3	2
Totals	25	16	17	16	21	15
(1 is the lowest/worst score, 5 is the highest/best score)						

Figure D.47: Product Concept Decision Matrix

The criteria used were chosen to ensure that the product was saving time and a viable product for the consumers and for production.

7. Cost:
 - a. Making sure the product is affordable to users by minimizing the material cost needed to build the product without sacrificing its quality
8. Ease of Use:
 - a. The product should be simple and intuitive to use, making it appealing and encouraging for the user to use.
9. Buildable:
 - a. The design must be realistic for the team to build and capable of being tested multiple times
10. Effectiveness:
 - a. The product should effectively solve the problem by reducing the amount of time and effort it takes to pick up leaves.
11. Differ from Market:
 - a. The product should stand out from current solutions in the market
12. Storage in Garage:
 - a. The product should be compact and space-efficient

Appendix E. Survey Questions

What is the total time it takes for you to pick up all the leaves in your yard and dispose them?

67 responses

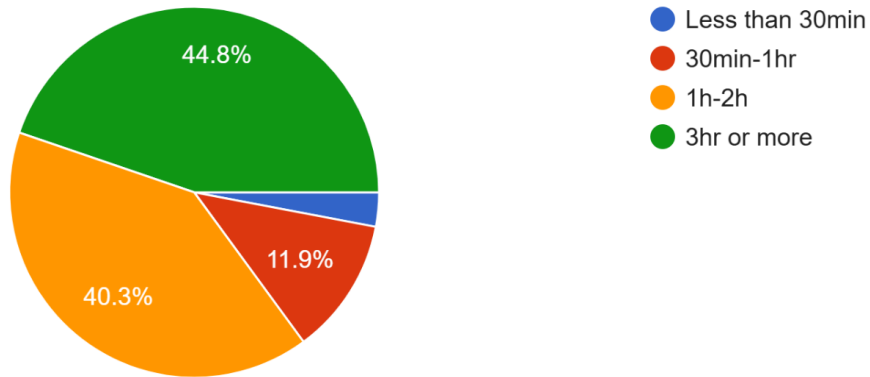


Figure D.48 Survey: Total time to pick up leaves in the fall

Product 1: Leaf Gulper

How much would this product impact your efficiency in picking up leaves?

67 responses

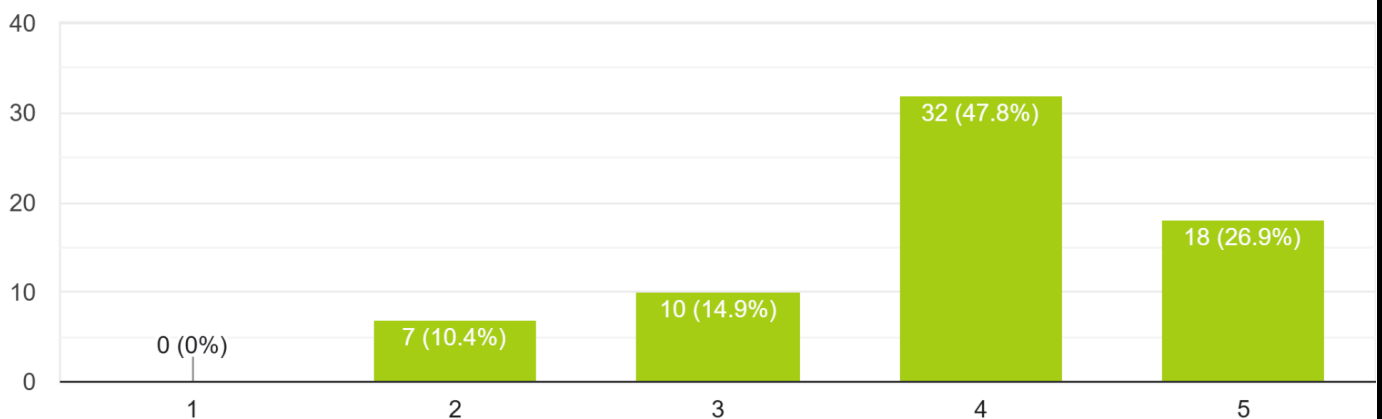


Figure D.49: Leaf Gulper product efficiency

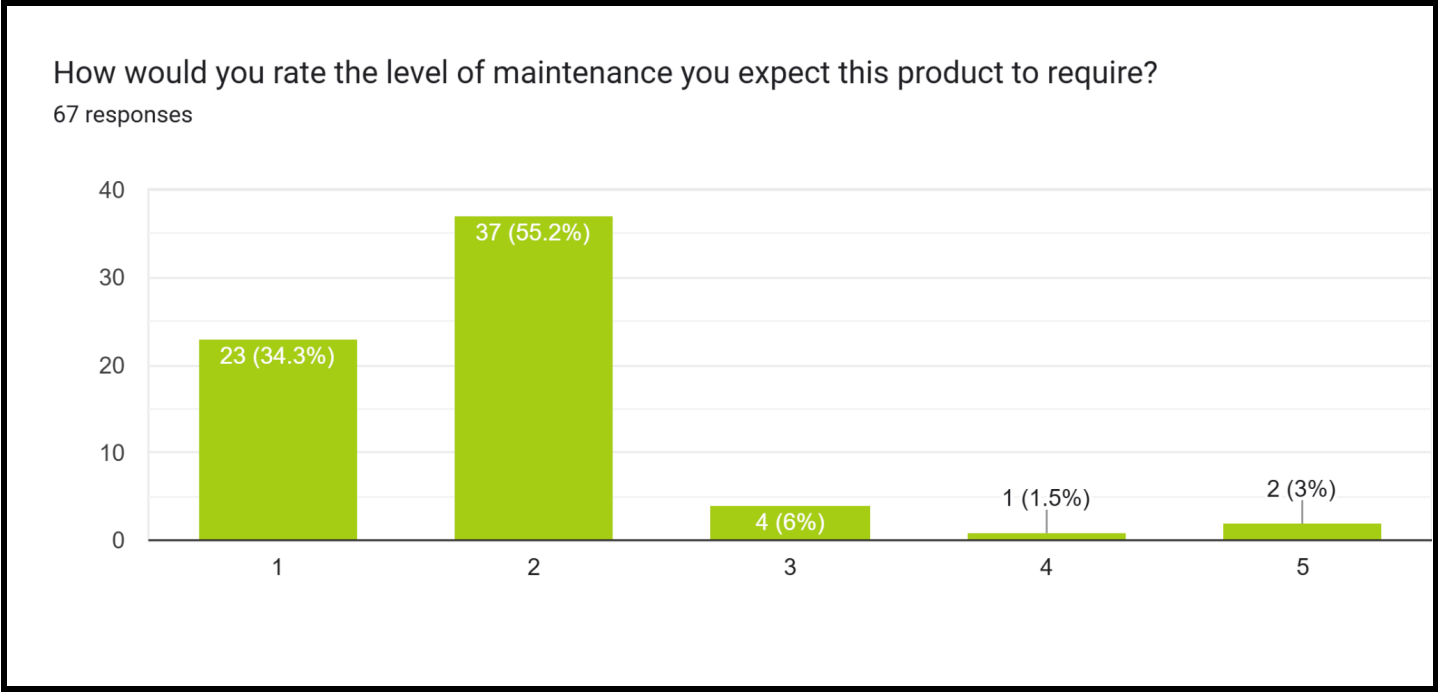


Figure D.50: Leaf Gulper maintenance

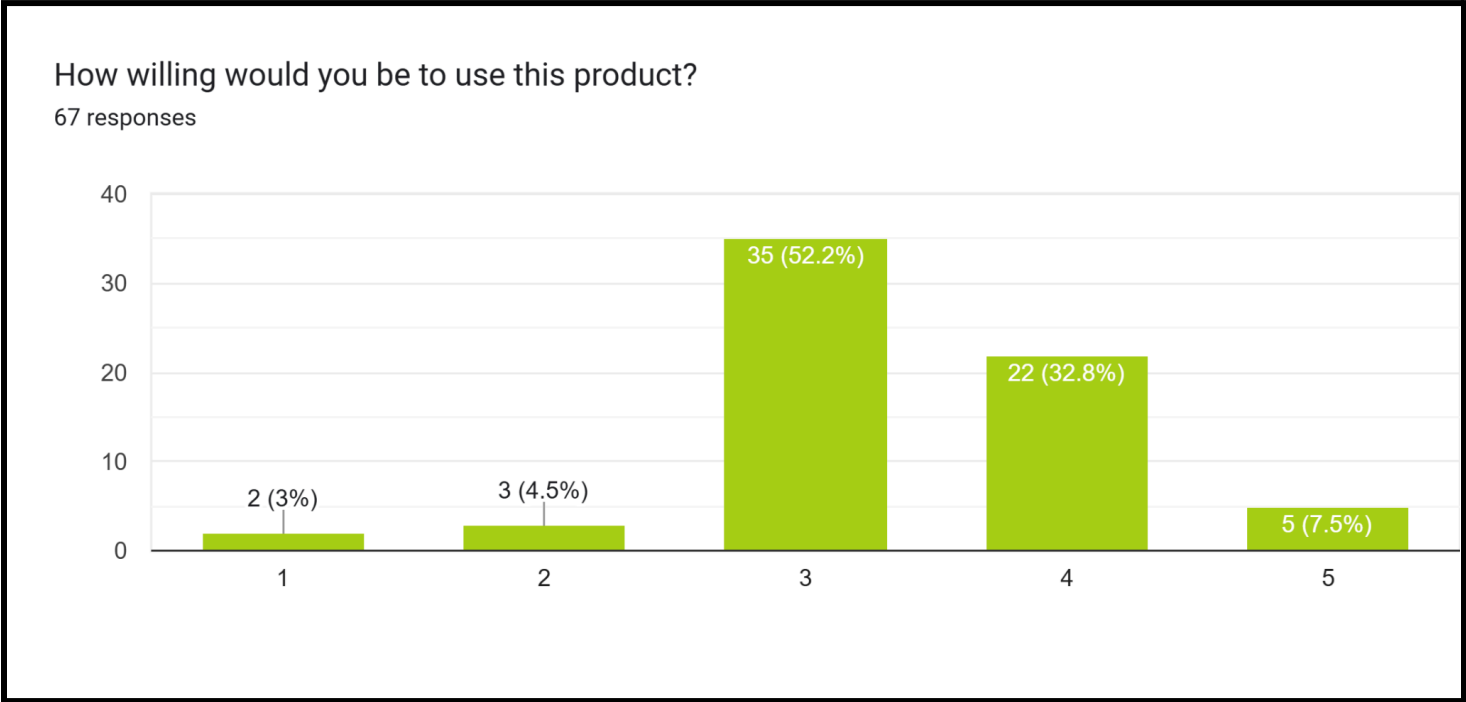


Figure D.51: Leaf Gulper Willingness

Product 2: Lawn Sweeper

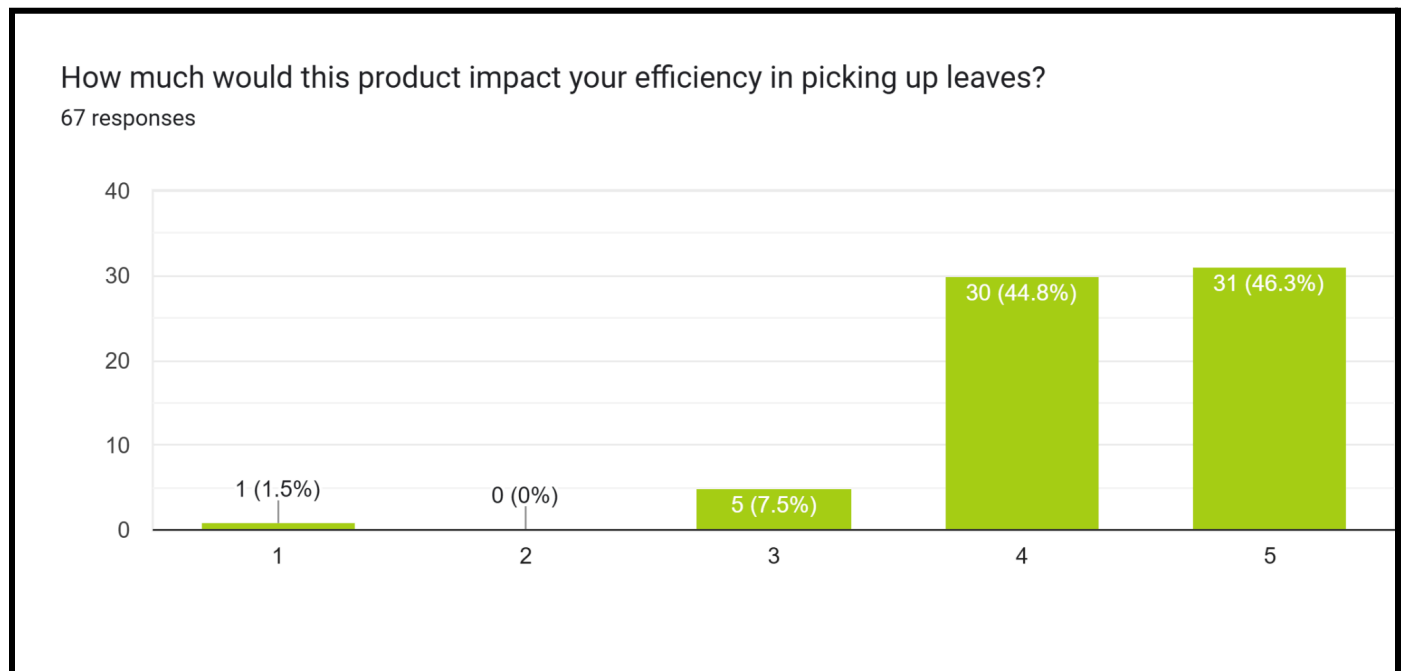


Figure D.52 Lawn Sweeper Efficiency

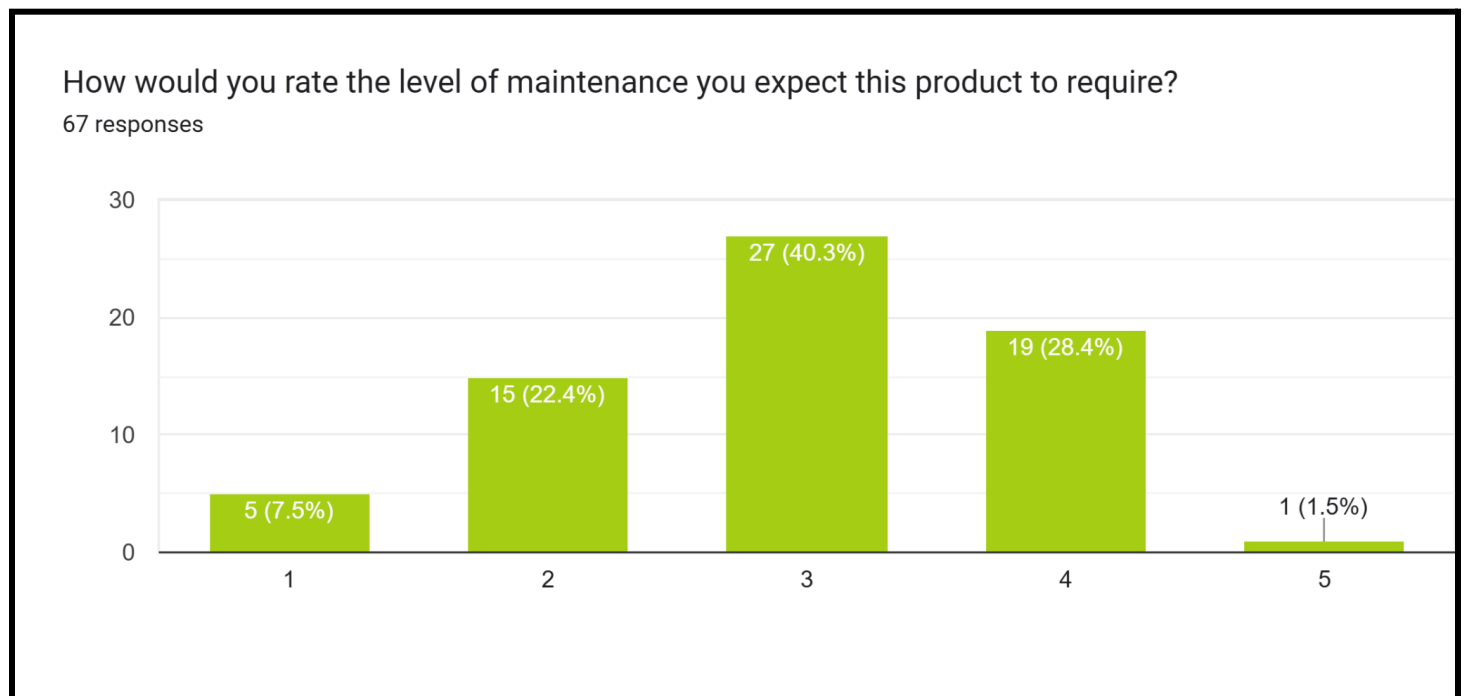


Figure D.53: Lawn Sweeper Maintenance

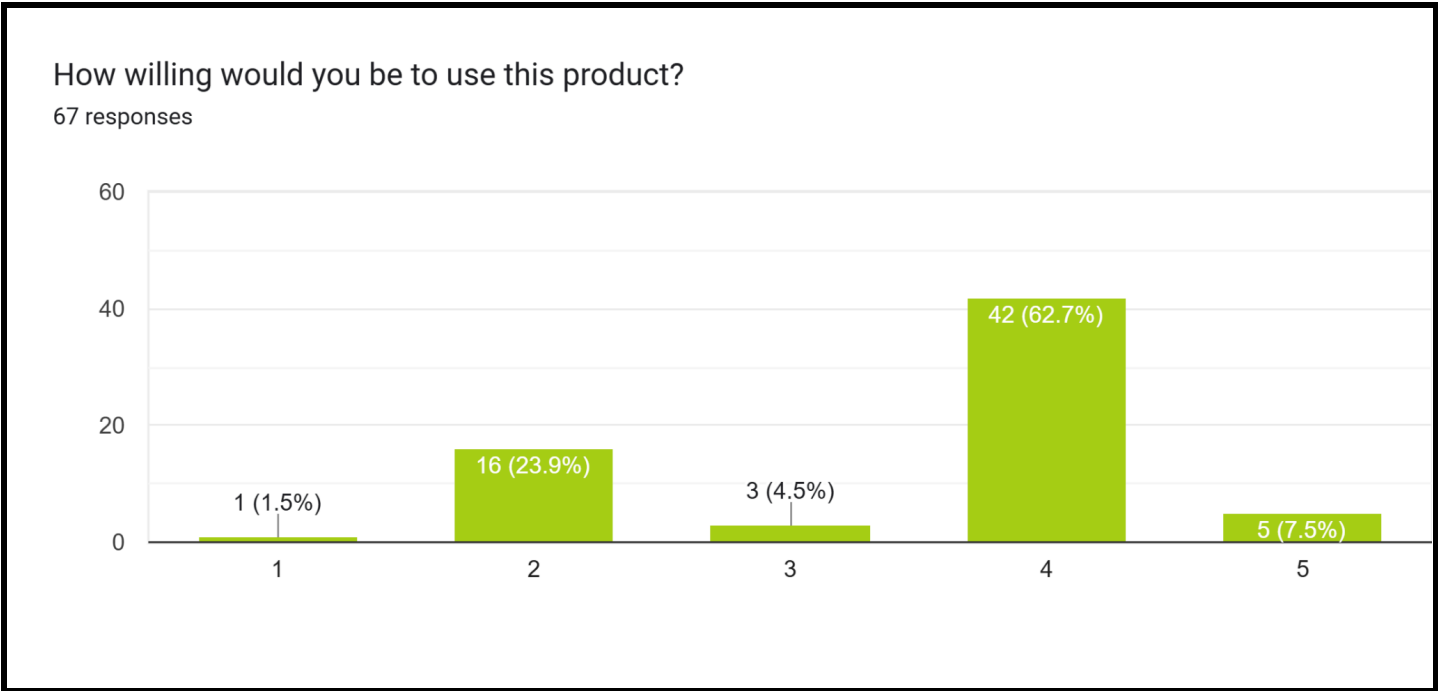


Figure D.54: Lawn Sweeper Willingness

Product #3: Auto Fridge Cleaner

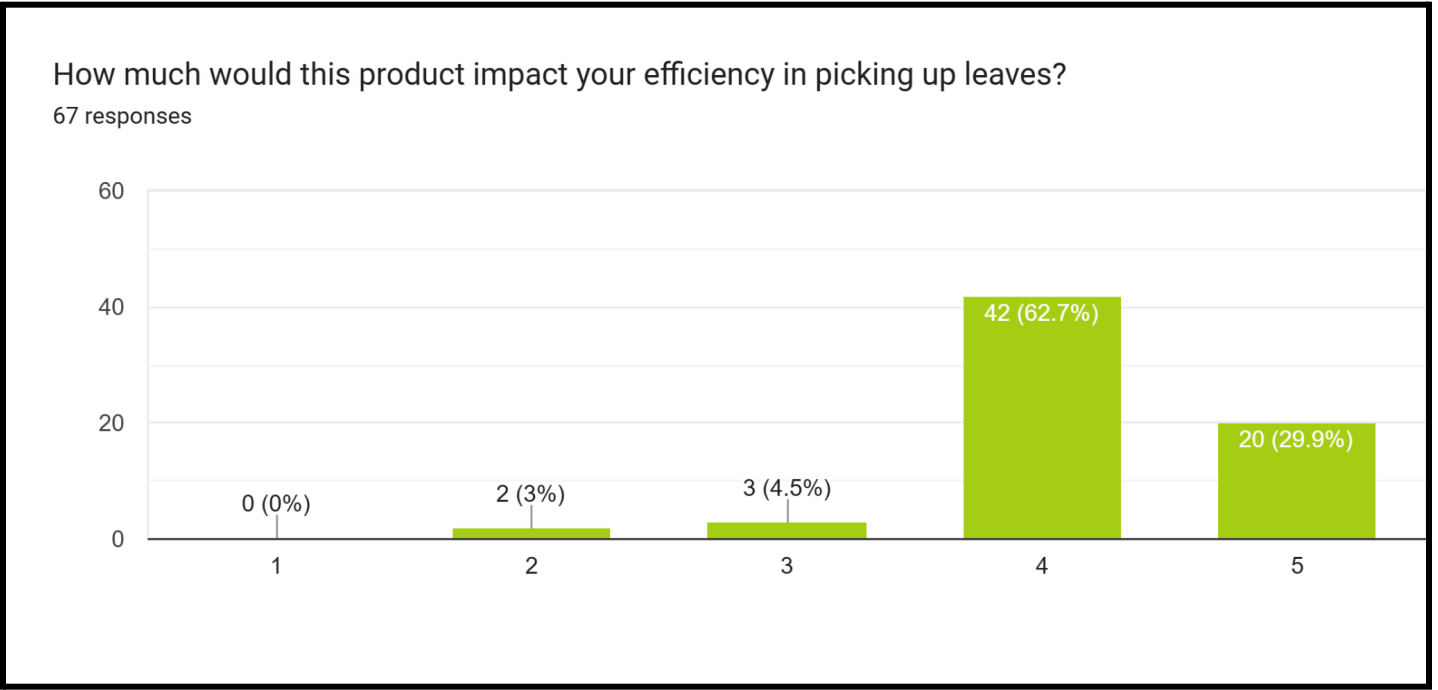


Figure D.55: Auto Fridge Cleaner Efficiency

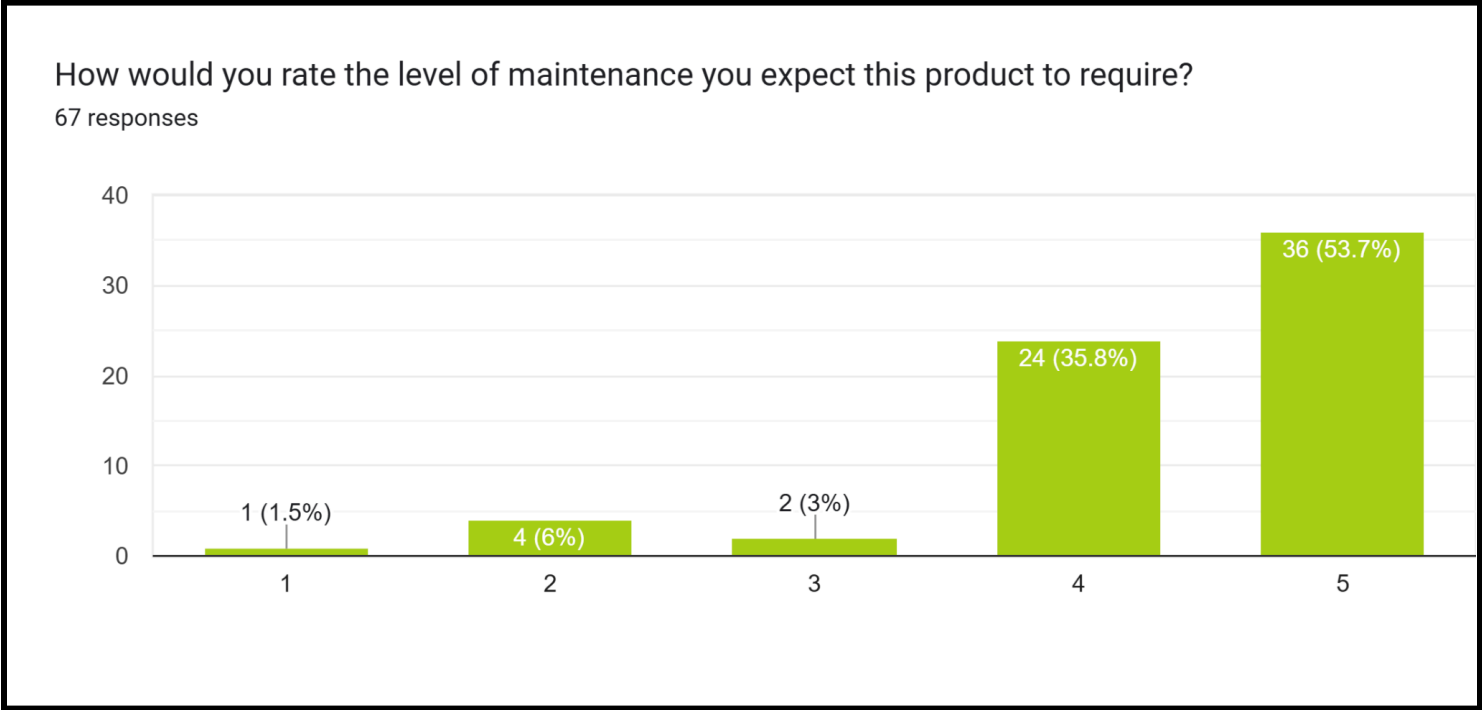


Figure D.56: Auto Fridge Maintenance

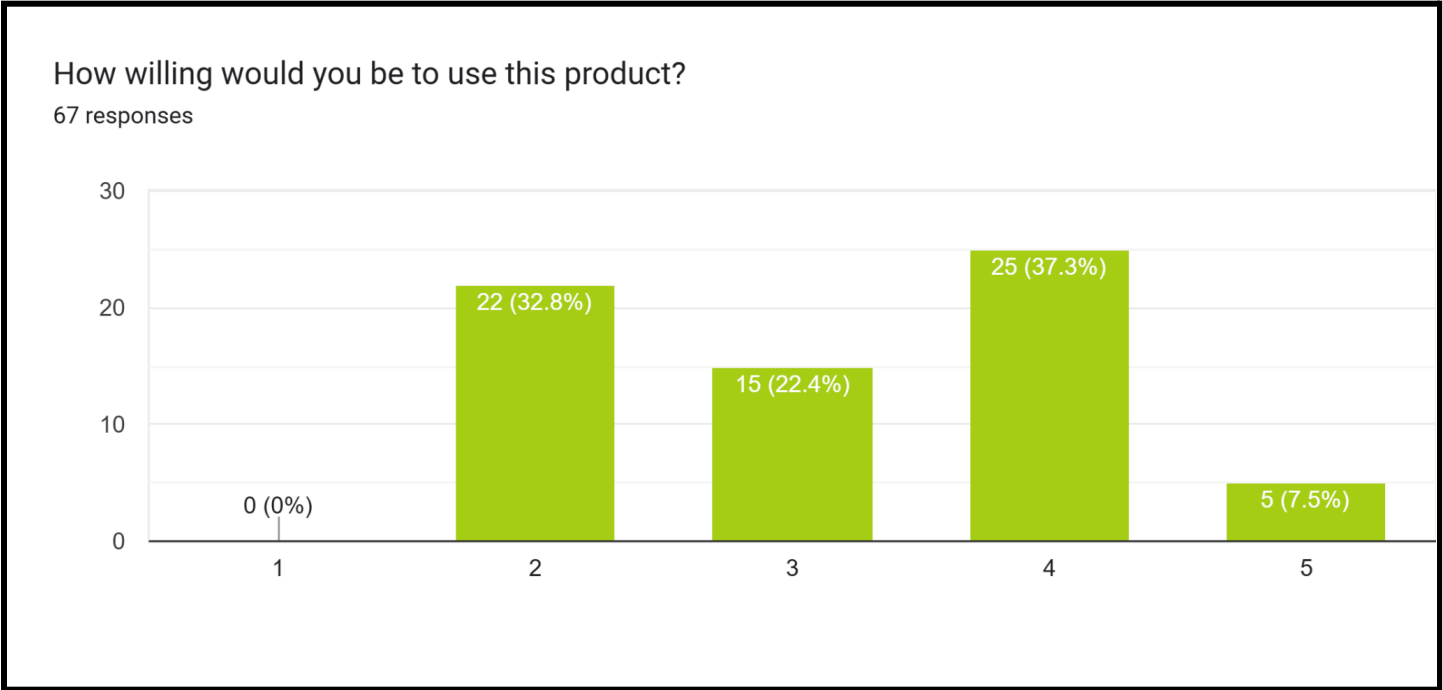


Figure D.57: Auto Fridge Willingness

How much would you be willing to spend on a product that solves this problem?

67 responses

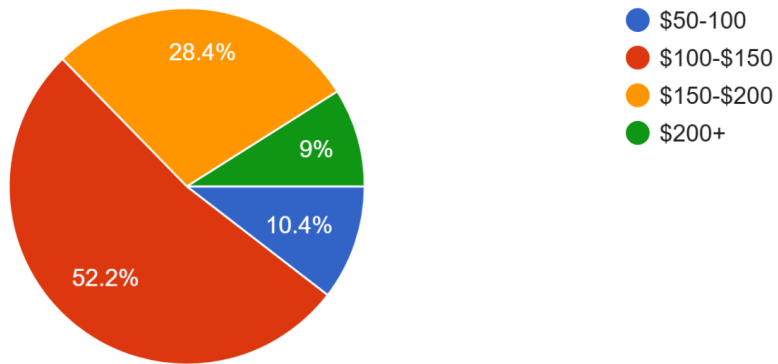


Figure D.58: Product Pricing

Which type of design would you prefer in a product solving this problem

67 responses

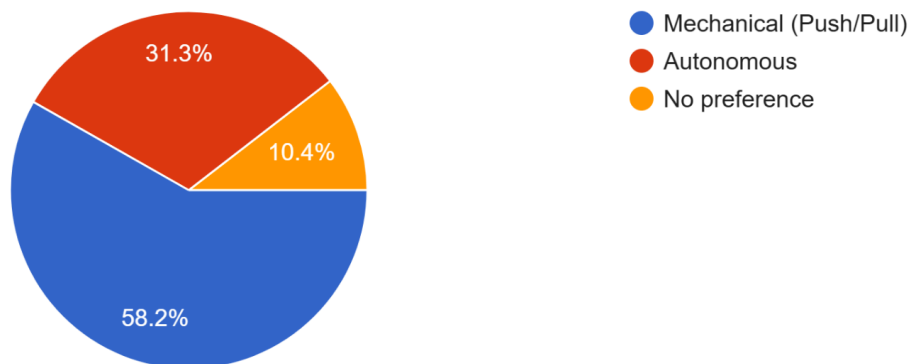


Figure D.59: Product Auto/Mechanical

Appendix F. Product Drawings

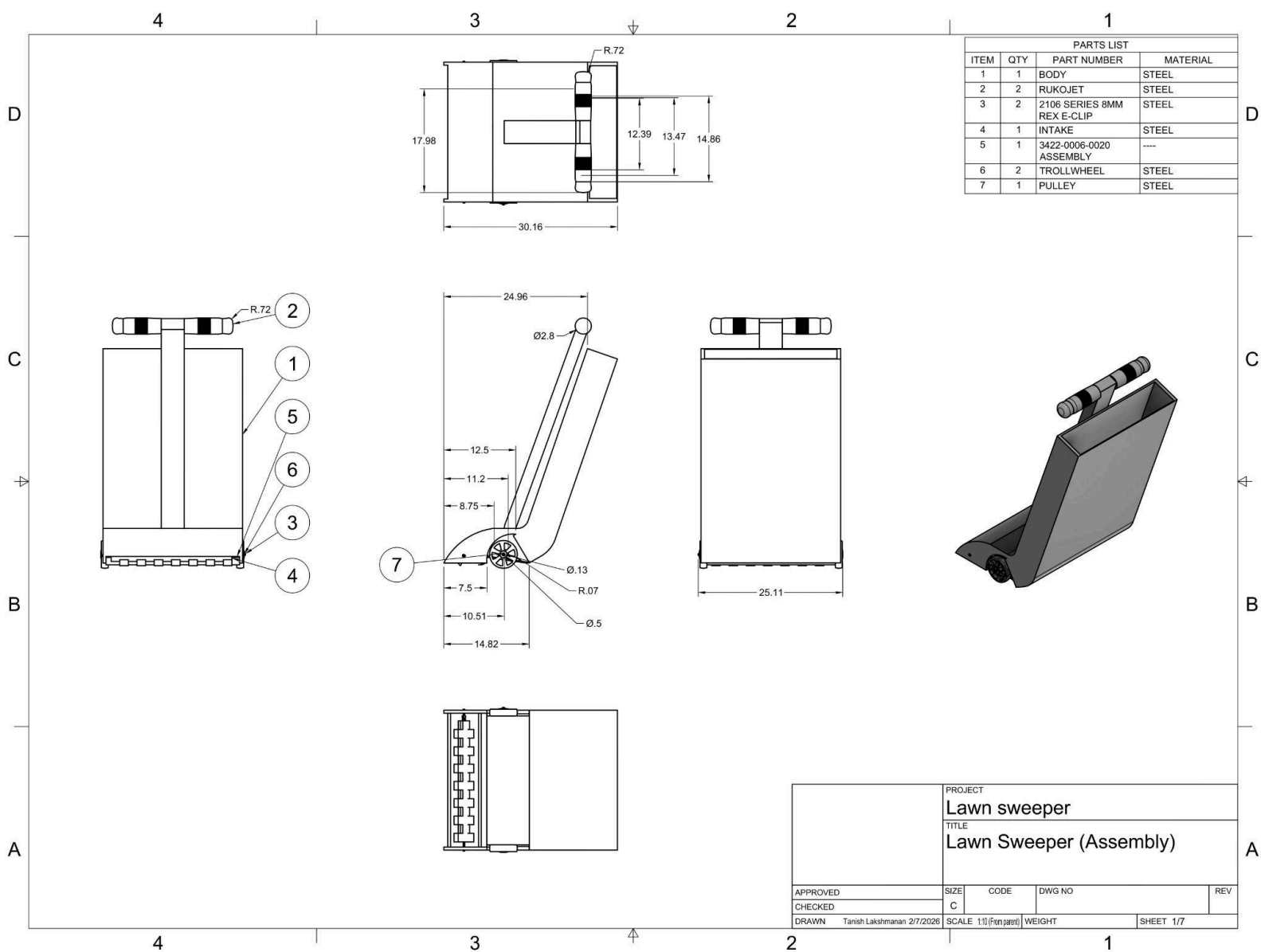


Figure D.60: Isometric and main views

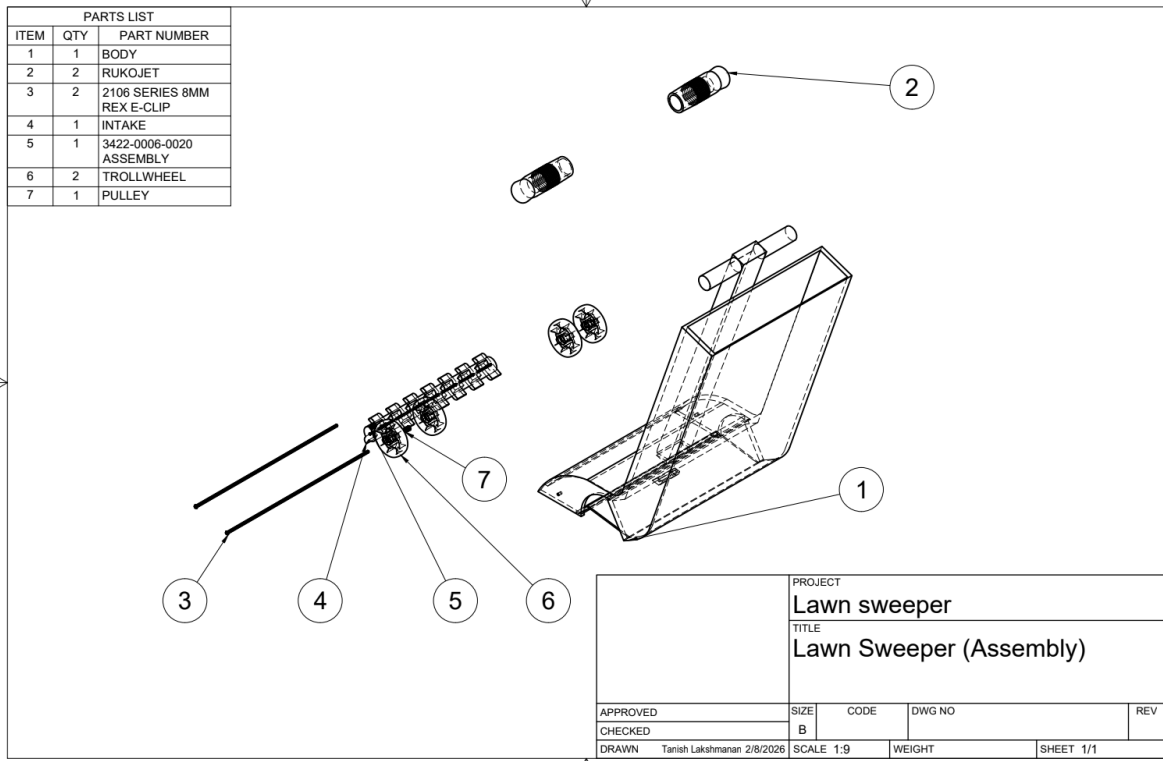


Figure D.61: Exploded isometric view

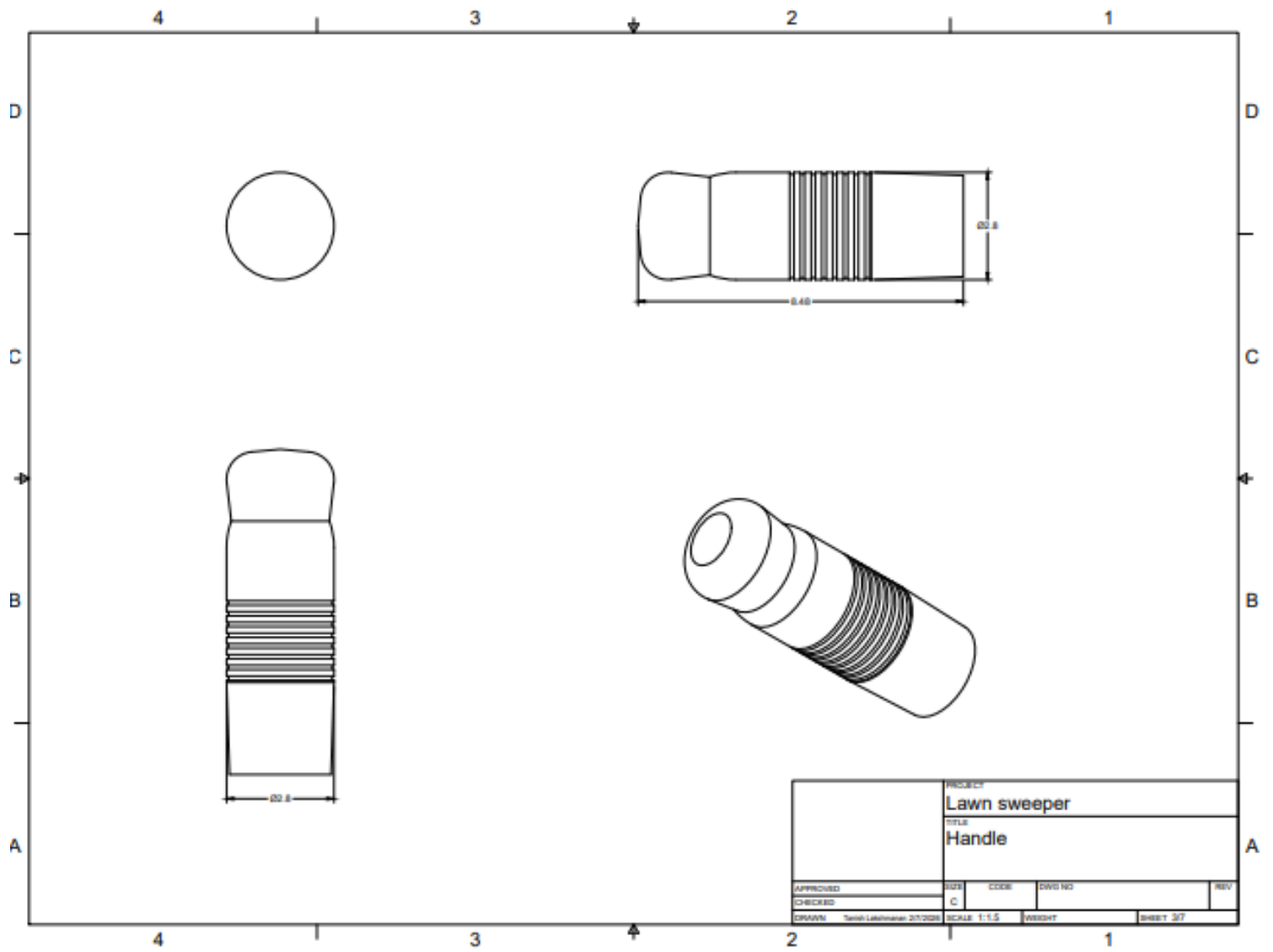


Figure D.62: Lawn sweeper handle

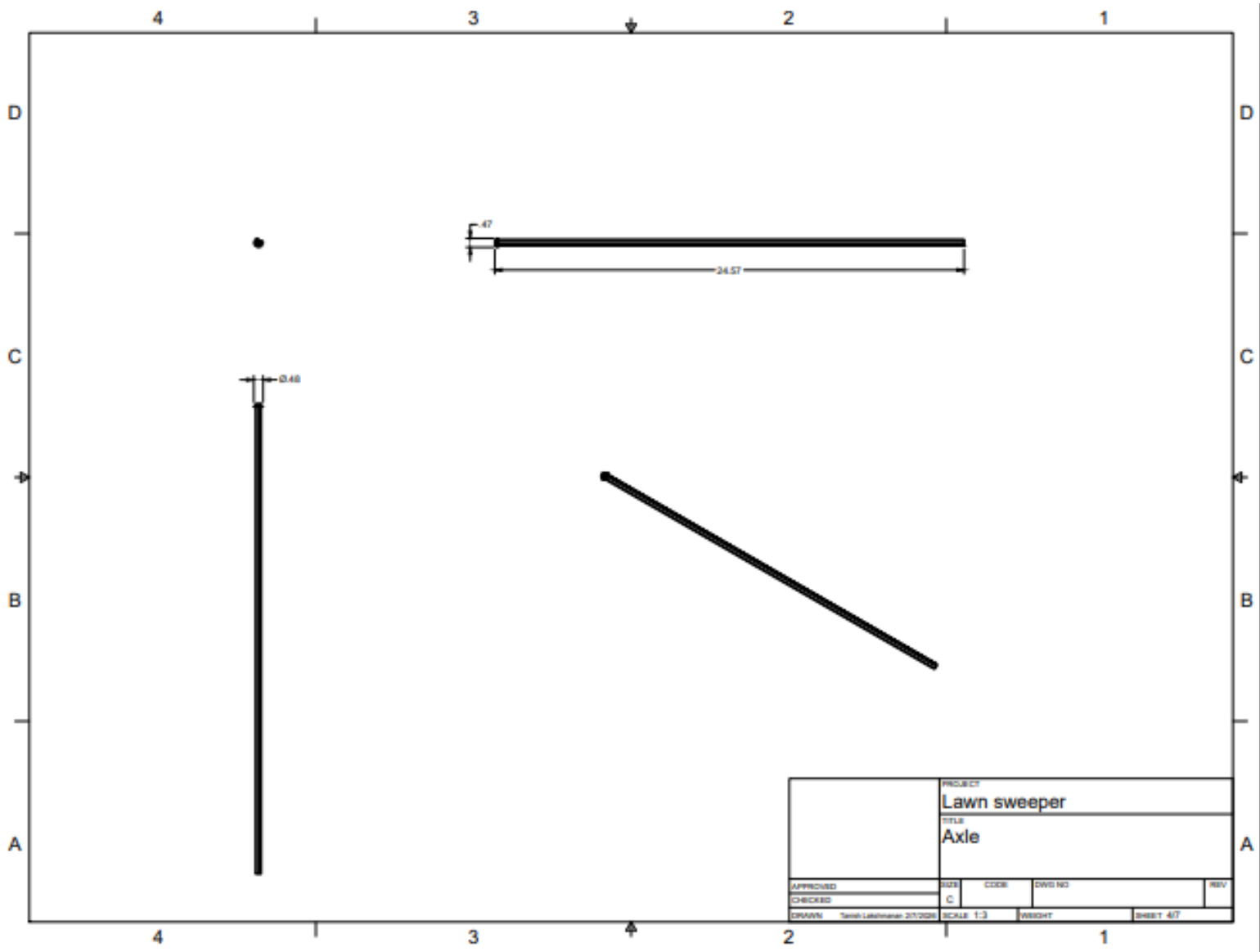


Figure D.63: Lawn sweeper axle

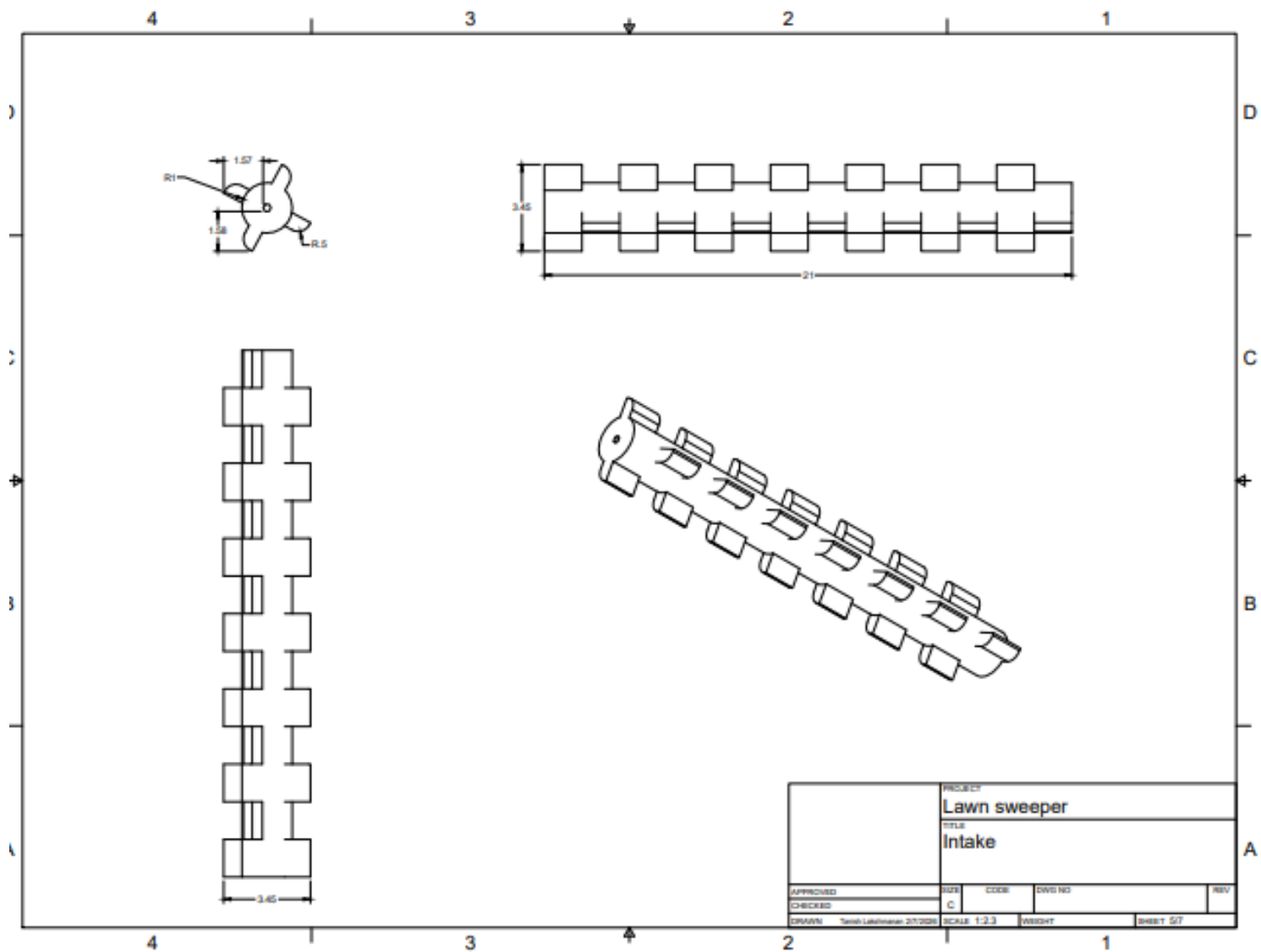


Figure D.64: Lawn sweeper intake

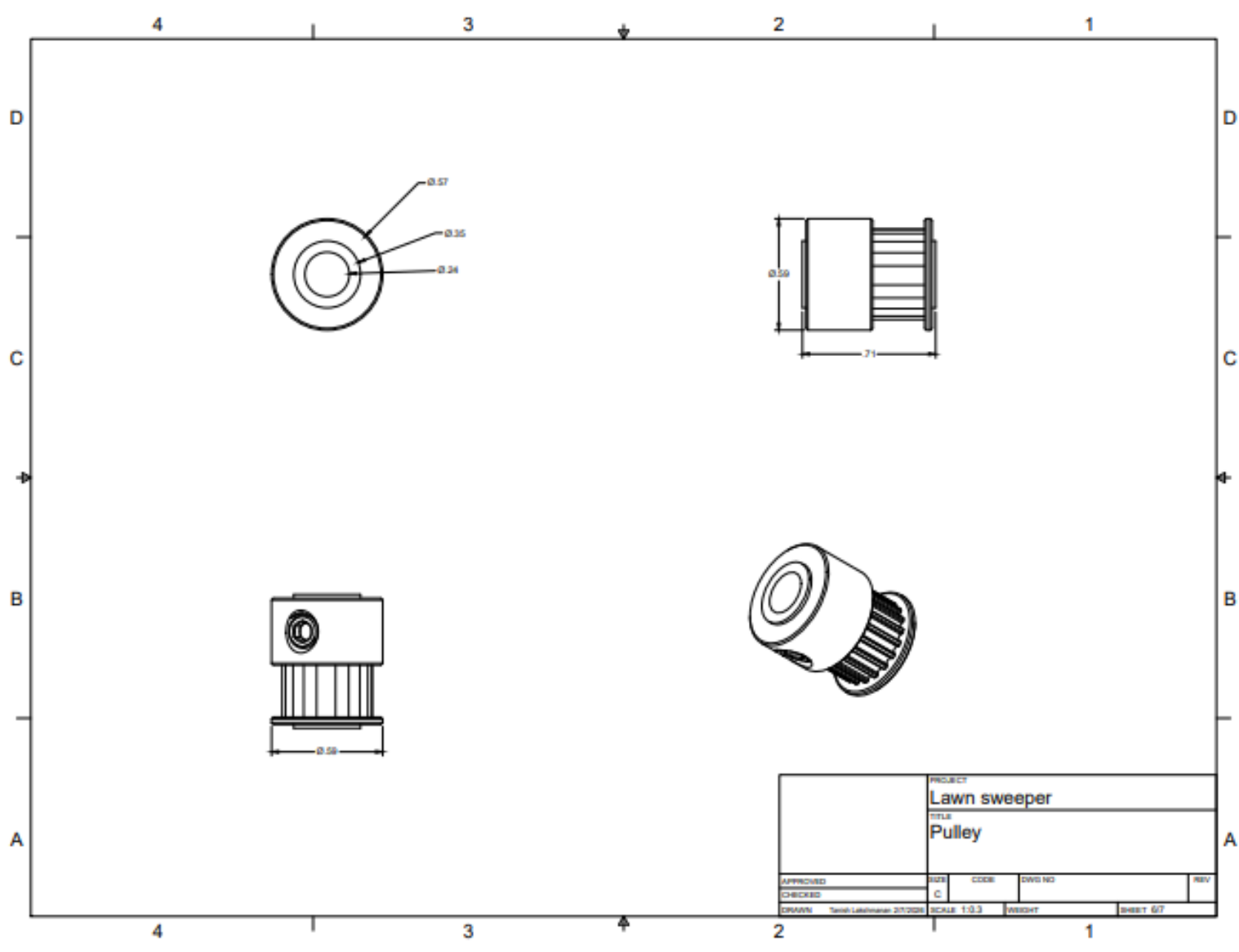


Figure D.65: Lawn sweeper pulley

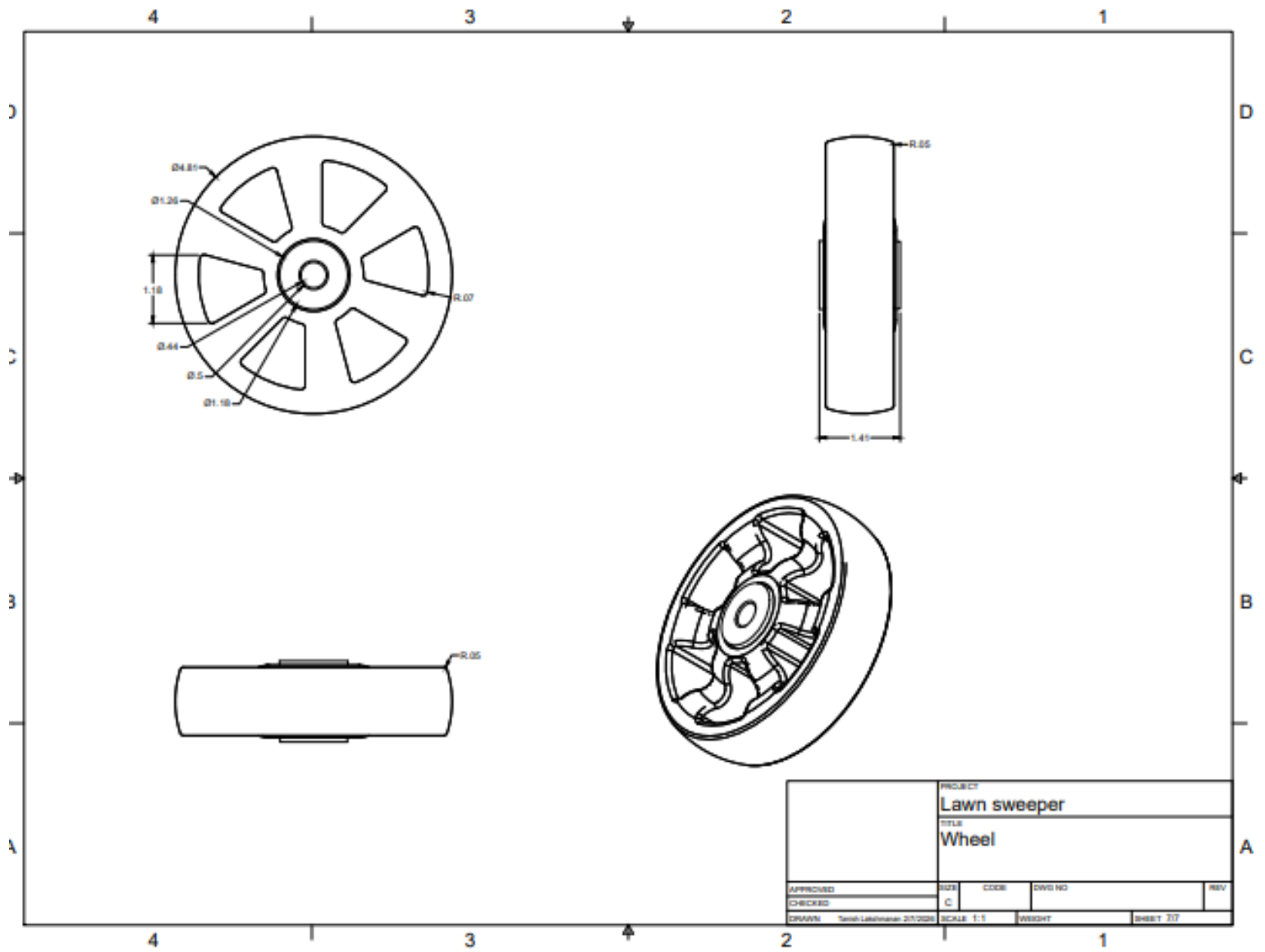


Figure D.66: Lawn sweeper wheel

PARTS LIST			
ITEM	QTY	PART NUMBER	MATERIAL
1	1	BODY	STEEL
2	2	RUKOJET	STEEL
3	2	2106 SERIES 8MM REX E-CLIP	STEEL
4	1	INTAKE	STEEL
5	1	3422-0006-0020 ASSEMBLY	----
6	2	TROLLWHEEL	STEEL
7	1	PULLEY	STEEL

Figure D.67: Parts list